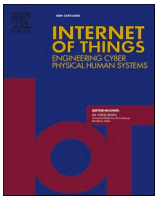
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# MULTI-BLOCK: A novel ML-based intrusion detection framework for SDN-enabled IoT networks using new pyramidal structure

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ABSTRACT

The ever-expanding Internet of Things (IoT) landscape faces significant security challenges due to limitations in traffic monitoring, device heterogeneity, and weak security practices, leaving networks vulnerable to multi-target and coordinated attacks like large-scale botnets and Distributed Denial-of-Service (DDoS). Existing intrusion detection systems can be computationally expensive and resource-intensive. This can be problematic for resource-constrained IoT devices with limited processing power, memory, and battery life. This paper proposes MULTI-BLOCK, a novel, multi-module framework that leverages machine learning, stateful P4 processing, and a Software-Defined Networking (SDN)-based multi-controller architecture. MULTI-BLOCK tackles critical management tasks within IoT networks, including synchronized communication, traffic monitoring, intrusion detection, and attack mitigation. This comprehensive framework comprises four modules. The first, the proposed pyramidal conceptually decentralized multi-controller structure (PCDMCS), introduces Decentralized Control Interfaces (DCIs) for real-time threat identification through a Decentralized Warning Conduit (DWC). The second module provides comprehensive network monitoring using P4-enabled 24-state tables. The third module leverages 30 new P4-extracted/calculated features and the Enhanced Weighted Ensemble Algorithm (EWEA) for enhanced anomaly detection. The final module presents a novel mitigation approach with 22 steps distributed across multiple controllers for scalability. Extensive evaluation using established IoT datasets (X-IIoTID, TON_IoT, and Edge-IIoTset) and diverse test scenarios (including single-victim and multi-victim attacks) demonstrates MULTI-BLOCK's exceptional performance, achieving high accuracy (99.75 %), precision (99.32 %), F1-score (99.53 %), recall (99.67 %), specificity (99.60 %), low false positive rates (FPR) (0.40 %), and low Average Detection Time (ADT) (2.11 ms). By offering a robust and adaptable solution against evolving threats, MULTI-BLOCK represents a significant advancement in IoT network security.

## 1. Introduction

The pervasive integration of the Internet of Things (IoT) into our daily lives has introduced unparalleled levels of convenience and efficiency. According to a recent report by Gartner, there will be an estimated 25 billion connected devices globally by 2025. However,

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this interconnectivity comes at the cost of increased vulnerability to intrusion attacks, presenting a concerning landscape of potential threats. The very essence of IoT lies in its intricate web of interconnected devices, spanning from smart appliances to critical infrastructure systems [1]. This intricate network, while promising enhanced automation and seamless communication, also opens up an expanded attack surface for malicious actors [2]. Intrusion danger within the IoT ecosystem is a formidable challenge, primarily due to the heterogeneity of devices, varying protocols, and often inadequate security measures [3]. Each device, regardless of its complexity, from resource-constrained wearables and sensors like smartwatches and fitness trackers to industrial control systems managing critical infrastructure, is a potential entry point for attackers to breach the system [4,5]. The consequences of such intrusions can range from data theft, privacy breaches, and device manipulation to more severe scenarios involving compromised critical infrastructure, like energy grids or healthcare systems. For instance, data theft could expose personal information such as social security numbers or credit card details. Privacy breaches could allow unauthorized access to camera footage from home security systems or baby monitors. Device manipulation could alter thermostat settings in a smart home, causing discomfort or even disrupting critical operations in a medical device within a hospital network.

Furthermore, compromised IoT devices can be harnessed as tools within larger botnets, amplifying the reach and impact of attacks, including instances like Distributed Denial of Service (DDoS) attacks [6,7]. This lack of uniform security architecture leaves gaps that attackers can exploit [8]. Moreover, the resource-constrained nature of many IoT devices often means that traditional security measures, such as robust encryption or complex authentication protocols, may be impractical to implement [9,10]. For example, robust encryption with complex key management schemes or computationally expensive authentication protocols may be too resource-intensive for devices with limited processing power, memory, and battery life constraints. To mitigate these escalating intrusion dangers, robust security frameworks tailored to the unique characteristics of IoT ecosystems are imperative [11].

Software-Defined Networking (SDN) has arisen as a prospective framework for overseeing and safeguarding expansive IoT networks [12]. Its centralized governance empowers network managers to automatically distribute network assets and deploy security measures for the identification and mitigation of attacks [13]. However, the standard SDN structure suffers from limitations such as a Single Point of Failure (SPOF) owing to the centralization of intelligence in a single controller, which also hinders scalability and performance [14]. To address this, researchers have explored various multi-controller architectures: hierarchical structures with global and regional controllers for balanced control or pyramidal structures (like ours) with a central controller coordinating regional controllers that manage local device-level controllers. This approach offers a trade-off between scalability and centralized management, with flat architectures being an alternative that distributes control evenly but increases complexity. The choice of architecture depends on specific network needs like scalability, fault tolerance, and real-time performance. This paper explores the potential of a pyramidal multi-controller architecture for enhanced intrusion detection in IoT networks. While SDN architecture offers several advantages, OpenFlow switches' low intelligence—which uses a stateless approach to packet processing—means that scalability and performance problems still exist [15,16]. Significant overhead for communications among both data and control plane in OpenFlow switches since they mostly rely on the controller's capabilities for network traffic routing and monitoring [17]. Furthermore, the OpenFlow version-based fixed behavior of these actions makes it difficult to modify header data and actions to satisfy different application needs [18,19].

To get around these restrictions, scientists developed a domain-specific language called P4 (Programming Protocol-independent Packet Processors) that enables data plane programming [20]. P4 makes it possible to define various header structures and the functions that go along with them for matching and specifying what the switch may do with each packet. P4 provides adaptability by offering a configurable pipeline that surpasses OpenFlow switches, empowering network administrators to customize packet processing based on application needs [21,22].

In recent years, researchers have leveraged machine learning (ML) techniques like supervised learning algorithms, specifically decision trees (DT), Support Vector Machines (SVMs), and deep learning models like Convolutional Neural Networks (CNNs) for identifying intrusion attacks in Software-Defined Internet of Things (SD-IoT) networks [23]. ML-enabled methods have proven effective in identifying these types of attacks. Nevertheless, training the framework on normal and malicious behavior is necessary to build a successful ML model [24], which requires selecting the most relevant features from the dataset [25]. Feature engineering plays a critical role in constructing an efficient detection framework as it helps in reducing cost and time complexity [26]. Therefore, focusing on packet feature engineering is crucial for the success of intrusion detection in SD-IoT networks [27].

### 1.1. Problem statement and motivation

The pervasive integration of Internet of Things (IoT) devices into our daily lives has revolutionized various aspects of our personal and professional environments. However, this interconnectedness introduces significant security vulnerabilities, making robust intrusion detection mechanisms crucial. This paper focuses on addressing the following key challenges within the realm of IoT security:

- **Effective Traffic Monitoring with Reduced Overhead:** Traditional traffic monitoring techniques often generate significant network overhead due to frequent synchronization and communication between network devices. This overhead can be particularly detrimental in resource-constrained IoT networks. Our research aims to develop a lightweight traffic monitoring approach that minimizes communication overhead while maintaining effective network visibility for intrusion detection purposes.
- **Intrusion Detection in Heterogeneous IoT Environments:** The diverse nature of IoT devices, encompassing wearables, sensors, industrial control systems, and smart home appliances, presents a significant challenge for intrusion detection. These devices often have varying communication protocols, limited processing power, and inadequate security measures, making them susceptible to a

wide range of attacks. Our proposed solution aims to be adaptable and effective in identifying intrusions across this heterogeneous device landscape.

- **Combating Coordinated Multi-Target Attacks:** Current intrusion detection systems often struggle to identify and mitigate coordinated attacks that target multiple devices simultaneously. This can be particularly concerning in scenarios involving botnets, where compromised IoT devices are used to launch large-scale Denial-of-Service (DoS) attacks. Our research focuses on developing an intrusion detection system that can effectively detect and respond to coordinated multi-target attacks in real-time.
- **Real-World Data-Driven Intrusion Detection Models:** The scarcity of research utilizing real-world IoT network traffic data hinders the development of practical intrusion detection systems. Many existing studies rely on synthetic or simulated data, which may not accurately reflect real-world network behavior and attack patterns. Our research leverages real-world IoT network traffic data to train and evaluate our intrusion detection model, ensuring its effectiveness and applicability to real-world scenarios.
- **Resource-Efficient Intrusion Detection for IoT Networks:** Existing intrusion detection systems can be computationally expensive and resource-intensive. This can be problematic for resource-constrained IoT devices with limited processing power, memory, and battery life. Our research prioritizes developing a lightweight and efficient intrusion detection system that can operate effectively within the resource limitations of typical IoT devices.

By addressing these specific challenges, our research aims to contribute to the development of a robust and practical intrusion detection system tailored to the unique security needs of IoT networks. This system will provide efficient traffic monitoring with minimal overhead, effectively detect intrusions across heterogeneous devices, identify and mitigate coordinated multi-target attacks, leverage real-world data for improved accuracy, and operate efficiently within resource-constrained IoT environments.

### *1.2. Research contribution*

This research tackles multifaceted challenges in IoT network intrusion detection by proposing a novel, integrated framework. Our work significantly contributes to the field through the following key aspects:

- **Integrated Framework for Enhanced Intrusion Detection:** Our research introduces a novel, integrated architecture that combines state-driven P4 technology for data plane control, machine learning for intrusion pattern identification, and a pyramidal, conceptually decentralized multi-controller structure (PCDMCS) based on SDN for enhanced scalability, fault tolerance, and real-time decision-making. This integrated approach surpasses existing solutions by offering a comprehensive and adaptable framework for intrusion detection within the complex and diverse environment of IoT networks.
- **Expanded Feature Set for Diverse Attack Detection:** The framework introduces a significantly enhanced feature set encompassing 30 new features. This includes 16 calculated features and 14 features directly extracted using P4 technology. This comprehensive feature set empowers the framework to effectively identify a wider range of intrusion attack types (13 in this study) compared to prior works**.**
- **Innovative Communication Interfaces for Real-Time Response:** The PCDMCS module introduces two novel decentralized control interfaces (DCIs) – Decentralized General Interface (DGI) and Decentralized Alert Interface (DAI). These interfaces enable real-time attack detection and mitigation through the Decentralized Warning Conduit (DWC) and Decentralized Data Conduit (DDC). This innovative communication approach significantly enhances the system's responsiveness to threats compared to centralized architectures.
- **P4-enabled Decentralized Traffic Monitoring System (P4-DTMS):** This module presents the Sequential Multi-State Pipeline (SMSP) with 24 state tables for comprehensive network management. By leveraging P4 for efficient data plane programmability, P4-DTMS fosters improved accuracy and efficiency in intrusion detection and attack mitigation compared to traditional monitoring techniques.
- **Efficient Intrusion Detection and Prevention:** The P4-enabled Decentralized Intrusion Detection and Prevention System (P4-DIDPS) utilizes stateful P4 and machine learning to achieve robust intrusion detection and prevention. Notably, the framework employs the Enhanced Weighted Ensemble Algorithm (EWEA) for accurate identification of various intrusion attacks, contributing to the overall effectiveness of the system.
- **Advanced Decentralized Attack Mitigation Strategy:** The P4-enabled Decentralized Attack Mitigation Strategy (P4-DAMS) introduces a sophisticated attack mitigation strategy. This strategy features eight distinct phases and 22 sequential steps designed for the efficient handling of large-scale attacks across multiple controllers. This decentralized approach enhances the overall scalability and resilience of the system against coordinated assaults compared to centralized mitigation strategies.
- **Real-World IoT Traffic Feature Utilization:** A major strength of our framework lies in its utilization of authentic IoT traffic features for constructing the detection model. This approach stands in stark contrast to many existing studies that rely on simulated or non-actual network traffic data. By leveraging real-world data, our framework offers a more practical and generalizable solution for intrusion detection in real-world IoT network scenarios.

### *1.3. Beneficiaries of this research*

The advancements presented in this research on intrusion detection for SD-IoT networks hold significant value for a broad spectrum of stakeholders within the IoT ecosystem:

- **Network Security Professionals:** This research equips security professionals with a powerful and efficient intrusion detection framework specifically designed to address the unique challenges of safeguarding IoT networks. The framework's comprehensive approach, incorporating P4 technology, machine learning, and a decentralized SDN architecture, empowers them to proactively identify and mitigate evolving intrusion threats within these complex environments.
- **IoT Network Operators:** Network operators managing large-scale IoT deployments can leverage this framework to enhance the security posture of their networks significantly. The framework's robust intrusion detection capabilities provide operators with increased visibility into network activity, allowing them to detect and respond to potential security incidents promptly. This translates to a more secure and reliable network infrastructure for critical IoT applications.
- **IoT Device Developers:** By adopting the principles and best practices outlined in this research, IoT device developers can contribute to the creation of a more secure and trustworthy IoT ecosystem. The framework's focus on real-world traffic analysis and decentralized intrusion detection strategies can inform the design and development of IoT devices with improved intrinsic security features and built-in detection capabilities.
- **End-Users:** Ultimately, the benefits of this research trickle down to the end-users of IoT devices. The proposed framework contributes to a more secure and trustworthy connected environment, minimizing the risk of data breaches, privacy violations, and

**Table 1**
Surveying the landscape - summary of intrusion detection literature review.

| Ref | Year | Method | AI-based? | Strengths | Weaknesses |
|---|---|---|---|---|---|
| Ravi et al. [29] | 2020 | DDoS attack detection in IoT. | Yes | The detection accuracy rate is 96.28 %. | The tight dependence on a single categorization technique limits its usefulness. |
| Assis et al. [36] | 2020 | CNN-based system for DDoS detection in IoT | Yes | Threats are being stopped at the point of origin. | The CIC-2019 dataset was used for the evaluation, which is not suitable for IoT traffic characteristics. |
| V. Chang [38] | 2020 | Deep-IFS: Deep Learning model for IIoT intrusion detection | Yes | Deep-IFS achieves high accuracy (99.77 %) and scalability using the real-world IoT dataset BoT-IIoT. | Its supervised learning approach restricts learning from unlabeled data. |
| AlZahrani et al. [30] | 2021 | ML-based traffic monitoring and intrusion detection method in SDN. | Yes | They achieved a multi-class classification accuracy of 95.95 %. | Firstly, the detection capabilities are restricted to only four common attack types. Secondly, the study relies on the NSL-KDD dataset, which isn't specifically designed for SDN. |
| Saba et al. [32] | 2022 | An anomaly-based detection method using CNN. | Yes | Leverages the strengths of CNNs to analyze comprehensive IoT traffic data efficiently. | The model's accuracy drops significantly when evaluated on the BoT-IoT dataset compared to the NID dataset (99.51 % vs. 92.85 %). This raises concerns about the model's generalizability and effectiveness across diverse IoT environments and attack scenarios. |
| Ilango [35] | 2022 | Feed Forward CNN-based technique to detect DoS in IoT. | Yes | Efficient detection of low-rate DoS attacks in IoT and SDN environments. | The reliance on the CIC DoS 2017 dataset raises concerns about its generalizability. This dataset is not specifically designed for SDN or IoT environments, and the traffic features it contains may not accurately represent real-world SDN-IoT attack scenarios. |
| Long et al. [28] | 2022 | A statistical-based approach using packet payload entropy for intrusion attack detection. | No | Effective for simple attacks, such as DDoS. | Fails to address complex, multi-packet attacks. |
| Yosuf et al. [31] | 2022 | CNN-based anomaly detection method for SD-IoT networks. | Yes | OpenDayLight (ODL) is used as an appropriate SDN controller to detect DDoS assaults in IoT. | The RNN model was not accurately aligned with the complex properties of IoT network data after being trained using the data set NSL-KDD. |
| Naula et al. [34] | 2022 | A DRL-enabled technique for identifying low-rate DDoS assaults in SDN. | Yes | This strategy included elements involving monitoring of traffic, a flow sampling process, and an affordable intrusion prevention mechanism. | The technique falls short of addressing the complexities of scalability in SDN-based IoT networks and is incapable of detecting intricate attack patterns. |
| Gouda et al. [33] | 2023 | FMDADM: Multi-Layer ML-based techniques for intrusion detection in SD-IoT. | Yes | The integration of ML techniques with stateful packet processing to extract features and identify an assault on the data plane with high accuracy. | The constantly changing characteristics of IoT networks are not taken into account. |
| Khan et al. [37] | 2023 | AutoML: intrusion detection framework | Yes | The proposed framework exhibited high accuracy in detecting intrusions. | The datasets employed were not the most up-to-date. |
| Nandanwar et al. [39] | 2024 | AttackNet: A Deep learning enabled intrusion detection system for Industrial IoT environment. | Yes | A good DL model for IIoT botnet attack detection using an adaptive CNN-GRU architecture. | Reliance on labeled data and specific attack types can hinder the model's ability to adapt to new threats. As attack strategies evolve, the model's performance might deteriorate. |

device malfunctions caused by intrusion attacks. These foster a more reliable and secure user experience within the realm of the Internet of Things.

By addressing the critical needs of these diverse stakeholders, this research offers a valuable contribution to the advancement of intrusion detection in IoT networks, paving the way for a more secure and reliable future for the Internet of Things. The rest of the paper is organized as follows: Section 2 reviews the existing literature on SD-IoT network security; Section 3 provides a detailed explanation of the proposed framework; Section 4 analyzes the experimental results and the implications of the proposed framework; and Section 5 concludes the paper by providing possible directions for future studies.

## 2. Literature review

In the realm of safeguarding SD-IoT networks from malicious activities, the landscape has undergone significant expansion in recent times. The exploration of various strategies to detect and mitigate these threats has led to the emergence of diverse methodologies, each with distinct strengths and limitations. This section delves into a comprehensive discussion of prior works, aiming to provide a nuanced understanding of their contributions. Table 1 presents a summary of the literature review.

Several studies have explored statistical techniques for intrusion detection. Long et al. [28] proposed a method that analyzes packet payload entropy to distinguish normal traffic from malicious activity. However, this approach has limitations in SD-IoT networks. Intrusion attempts in these environments often involve coordinated attacks spread across multiple packets and exhibit stateful behavior. The reliance on individual packet entropy makes it difficult to detect such multi-packet, stateful attacks accurately.

Recent studies have focused on combining machine learning (ML) and deep learning (DL) techniques, either with or without the SDN paradigm, to improve the precision and efficacy of intrusion detection in IoT environments. One such approach is exemplified by LEDEM [29], a DDoS detection system designed specifically for SDN-enabled IoT networks. A comprehensive architecture was introduced by LEDEM for monitoring and analyzing network traffic to identify potential attacks. While LEDEM showcased promise, its reliance on a single classification method constrained its adaptability to diverse intrusion attack types, thereby limiting its effectiveness in comprehensive threat mitigation.

Similarly, the authors in [30] explored the application of ML algorithms within an SDN controller for network traffic monitoring and intrusion detection. Their work focused on utilizing various tree-based techniques like Decision Trees, Random Forests, and XGBoost for attack identification within a Network Intrusion Detection System (NIDS). The NSL-KDD dataset served as the foundation for training and testing their proposed method. By leveraging just a subset of the dataset's features (five out of 41), they achieved a multi-class classification accuracy of 95.95 %, successfully detecting attacks and classifying them into four different categories. However, this approach presents some limitations. Firstly, the detection capabilities are restricted to only four common attack types, neglecting the diverse range of attacks specifically targeting SDN or IoT. Secondly, the study relies on the NSL-KDD dataset, which isn't specifically designed for SDN environments. More specialized datasets are available for improved accuracy. Finally, the limited use of only five features raises concerns about under fitting. This could potentially hinder the training process and lead to insufficient learning.

Yousuf et al. [31] presented DALCNN, utilizing OpenDayLight (ODL) as an appropriate SDN controller for spotting DDoS assaults in IoT to fill the gap among ML approaches and the complexities of IoT network traffic. However, a significant drawback was found: the training of the Recurrent Neural Network (RNN) method on the NSL-KDD dataset did not perfectly match the complex features of IoT network traffic. Deep Neural Networks (DNNs) emerged as another avenue for enhancing intrusion attack detection. Another study proposed a CNN-based approach for anomaly detection in IoT [32]. This method leverages the strengths of CNNs to analyze comprehensive IoT traffic data efficiently. The authors claim that the model can identify various intrusion attempts and anomalous traffic patterns. However, the proposed method exhibits some limitations. Notably, the model's accuracy drops significantly when evaluated on the BoT-IoT dataset compared to the NID dataset (99.51 % vs. 92.85 %). This raises concerns about the model's generalizability and effectiveness across diverse IoT environments and attack scenarios.

Turning attention to a separate study [33], the authors integrated ML approaches with stateful packet processing to develop a complete framework for intrusion detection in SDN-IoT networks. This framework includes a special function called the Double-Check Mapping Function (DCMF). This function is designed to analyze data directly at the network layer (data plane) to extract important characteristics. These features were then examined using machine learning techniques for categorization. Using Deep Reinforcement Learning (DRL), a new method [34] was developed to identify DDoS attacks with low attack rates in SDN. This strategy included several elements involving flow sampling, monitoring of traffic, and an adaptable intrusion prevention system (IPS) for quick mitigation. Nevertheless, difficulties in resolving the subtleties of SDN-enabled IoT network scalability continued, highlighting the necessity for more improvements.

Similarly, another work introduced an AI-based anomaly detection system utilizing a FeedForward-Convolutional Neural Network (FFCNN) for identifying Low-Rate Denial-of-Service (LR DoS) attacks within IoT-SDN networks [35]. The study employed the CIC DoS 2017 dataset for training and evaluation. The authors implemented an iterative feature selection process using SVM to identify the most relevant features for attack detection. The FFCNN's performance surpassed various ML algorithms like J48, Random Forest, and Multi-Layer Perceptron (MLP). However, despite the good accuracy, the approach has limitations. First, the reliance on the CIC DoS 2017 dataset raises concerns about its generalizability. This dataset is not specifically designed for SDN or IoT environments, and the traffic features it contains may not accurately represent real-world SDN-IoT attack scenarios. Second, the detection capabilities are restricted to a single attack type – Low-Rate DoS. This limits the ability to assess the model's effectiveness against a broader range of intrusion attempts commonly targeting SDN-IoT network.

Different strategies were also investigated to improve the Deep Learning systems' ability to identify various intrusion assaults. Another research [36] suggested a CNN-based solution for anticipatorily stopping DDoS assaults at their origin when identifying them in IoT networks. Although this method showed encouraging findings, the CIC-DDoS2019 dataset review highlighted the necessity for careful examination of the dataset's compliance with the complexities of IoT network traffic dynamics. In a different direction, in [37], an AutoML intrusion prevention architecture was used to find the best-supervised classifiers, which led to the creation of an ensemble method using soft voting. The framework's results demonstrated a high degree of accuracy in identifying intrusion events. Nevertheless, it is crucial to recognize that the datasets employed may not have been the most current, and their comprehensiveness in encapsulating the entirety of attack intrusions may be limited. Furthermore, the framework's lack of consideration for stateful network cases warrants attention. The authors in [38] presented a significant work in this area. They proposed a Deep Learning model (Deep-IFS) for IIoT intrusion detection. Deep-IFS utilize LocalGRUs for local traffic patterns and a Multi-Head Attention layer to capture long-range dependencies. For scalability, Deep-IFS leverages fog computing, distributing training and processing across worker nodes. While the Deep-IFS offers advantages, the authors acknowledge limitations. Firstly, its supervised learning approach restricts learning from unlabeled data. Secondly, data privacy, crucial for sensitive industrial applications, remains unaddressed.

## 3. The multi-block framework proposal

The proposed framework consists of four modules. The first module proposed a pyramidal conceptually decentralized multi-controller structure (PCDMCS) and introduced the secure decentralized communication scheme (SDCS) for multi-controller communication. The SDCS Scheme introduces two distinct decentralized control interfaces (DCIs): the Decentralized General Interface (DGI) and the Decentralized Alert Interface (DAI). The two proposed interfaces facilitate real-time attack detection and mitigation, enabling early threat identification and response. The second module, Proposed P4-enabled Decentralized Traffic Monitoring System (P4-DTMS), serves as the core of the framework. It comprises 24 state tables that provide a comprehensive overview of managing the IoT network. The P4-DTMS module acts as input for other modules, such as intrusion detection and traffic mitigation, enhancing their accuracy and effectiveness. The third module, the Proposed P4-enabled Decentralized Intrusion Detection and Prevention System (P4-DIDPS), utilizes state-driven P4 and ML to detect and prevent intrusion attacks. By employing 30 P4-extracted and calculated features, the module applies the Enhanced Weighted Ensemble Algorithm (EWEA). The stateful nature of P4 enables detailed network visibility and timely identification of attacks at the data plane. To mitigate attacks effectively, the last module, Proposed P4-enabled Decentralized Attack Mitigation Strategy (P4-DAMS), presents a unique approach consisting of 22 steps. This decentralized strategy disperses the mitigation process among several controllers, enhancing scalability and reliability in handling larger-scale and coordinated attacks. Fig. 1 shows the workflow interaction among the different components in the proposed framework.

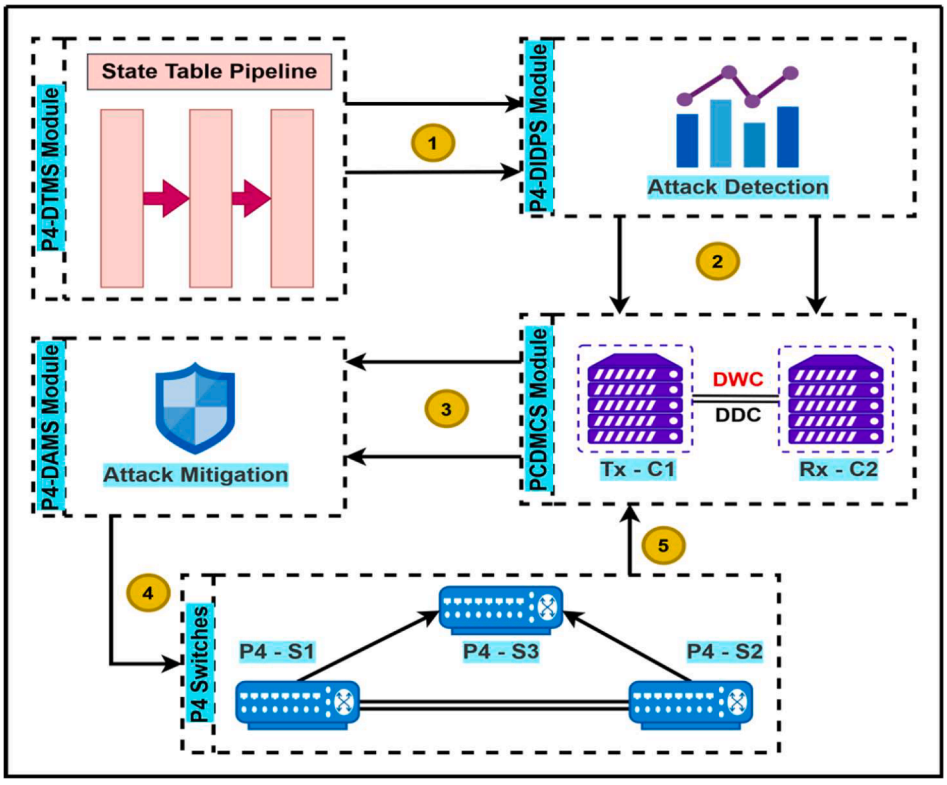



**Fig. 1.** Flow in harmony - workflow interaction among the components in the proposed framework.

Fig. 2 depicts the layered architecture of the proposed framework. Every layer contributes differently to improving the network's resilience and security. The MULTI-BLOCK architecture reduces the impact of intrusions on the whole network infrastructure by strengthening the defence against them through a mix of synchronization, monitoring, detection, and mitigation strategies.

The proposed MULTI-BLOCK framework incorporates a logically decentralized structure to address the complex network landscape of the Internet. With the Internet serving as a connection medium for multiple Internet Service Providers (ISP) networks, which, in turn, interconnect numerous Local Area Networks (LANs) comprising diverse nodes, the scale and volume of these LANs continue to grow rapidly. However, the proliferation of IoT nodes within these LANs introduces vulnerabilities to malicious software outbreaks, thus making them potential origins for various intrusion attacks. In an extreme scenario, when IoT nodes become infected and form botnets across multiple compromised ISP LANs, they pose a significant threat by launching attacks on remote servers.

These attacks can infiltrate LANs and manipulate compromised devices, potentially leading to severe consequences. Resource exhaustion on network controllers and servers can cripple vital functions, data exfiltration through compromised devices can expose sensitive information, and DoS attacks can render critical infrastructure inaccessible.

The MULTI-BLOCK architecture tackles these challenges by employing a multi-layered defence strategy within the LAN. Unlike traditional solutions overwhelmed by attack complexity, MULTI-BLOCK utilizes advanced intrusion detection techniques to identify botnet activity early before it causes widespread damage. By leveraging SDN, the framework implements granular traffic control measures. This isolates infected devices, preventing them from communicating with external servers and disrupting botnet operations. Additionally, containing attacks at the LAN level minimizes the burden on central controllers and servers, ensuring critical network resources remain available for legitimate activities. Ultimately, MULTI-BLOCK prioritizes the protection of controllers and distant nodes by confining malicious activities within the compromised LAN and preventing them from accessing the broader internet. This multi-layered approach safeguards vital network components, guaranteeing resource availability and integrity while mitigating potential damage and data breaches.

### 3.1. The first module (PCDMCS)

The initial module of the framework, denoted as the proposed pyramidal conceptually decentralized multi-controller structure (PCDMCS), is dedicated to the advancement of controller-to-controller (C2C) communications within SD-IoT networks. Its primary objectives encompass the reduction of communication overhead and the enhancement of intrusion attack detection. In pursuit of these aims, a secure decentralized communication scheme (SDCS) is introduced, which leverages a synergy of decentralized architecture and cryptographic methodologies to establish productive and secure C2C communications. The cardinal goals encompass the minimization of superfluous data exchange between controllers and the augmentation of intrusion attack detection capabilities through synchronized communication. The SDCS Scheme introduces two distinct decentralized control interfaces (DCIs): the Decentralized General Interface (DGI) and the Decentralized Alert Interface (DAI). Within the realm of C2C communication, the DGI interface encompasses the General Communication Mode (GCM), whereas the DAI interface encompasses the Alert Communication Mode (ACM). These interfaces collectively facilitate the real-time identification and containment of attack instances via two proposed Conduits: the Decentralized Warning Conduit (DWC) within the ACM context and the Decentralized Data Conduit (DDC) within the GCM mode. These Conduits collectively empower the expeditious recognition and response to emergent threats through the selective dissemination of pertinent notifications amongst participating controllers. By adopting this approach, the SDCS Scheme effectively curtails both data-control overhead and communication overhead among controllers, thereby optimizing resource utilization. The algorithmic representation of the SDCS Scheme is furnished within Algorithm 1.

Algorithm 1 initiates with the "Controller Identification and Authentication" phase. This initial phase necessitates the assignment of an exclusive identifier and a corresponding cryptographic key to each distinct controller. Preceding the commencement of

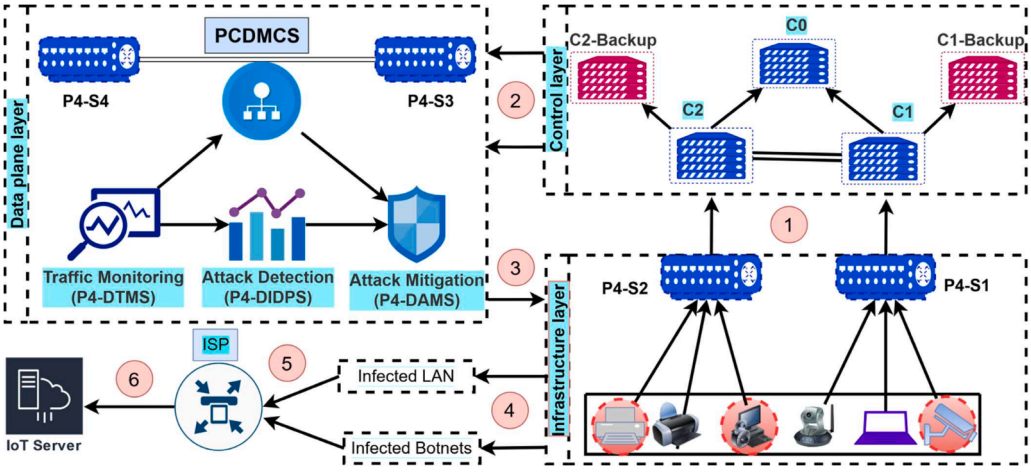



**Fig. 2.** The layered architecture of the proposed framework.

**Algorithm 1**

**Algorithm 1: The Proposed PCDMCS Algorithm**

**Inputs:** Controller Identifiers, Cryptographic Keys.
**Outputs:** Secure and Synchronized Communication among Controllers.

**Controller Identification and Authentication:**
**For each (Controller i) in the Network:**
    **Each** controller receives a unique identifier and a cryptographic key.
    **Authenticate** Controller i using its Cryptographic Key.
    **If (Authentication Successful):**
        **Establish** Secure Communication Conduit with Controller i.
    **End if**
**End For**

**Choose Communication Mode (GCM/ACM) based on Network Events**
**If (Selected Mode is ACM):**
    **Establish** a Decentralized Warning Conduit (DWC) linking the controllers.
    **Implement** the use of Message Queuing Telemetry Transport (MQTT) for the warning Conduit.
    **Identify** adjacent controllers.
    **Exchange** network data with adjacent controllers via the DWC.
**End if**
**Else If (Selected Mode is GCM):**
    **Establish** a Decentralized Data Conduit (DDC) linking the controllers.
    **Use** Message Queuing Telemetry Transport (MQTT) for the general Conduit.
    **Forward** regular network updates with other controllers through the DDC.
**End if**

**Key Exchange and Synchronization:**
    Controllers exchange cryptographic keys during initiation, enabling secure message handling.
    A secure synchronization mechanism ensures controllers possess up-to-date cryptographic keys.

**Intrusion Alert Propagation:**
**If (Intrusion Attack Detected):**
    **Encrypt** Intrusion Alert Message using DWC Key.
    **Transmit** Encrypted Alert through DWC of the ACM mode to other Controllers.
    **Broadcast** the notification across the IoT network.
**End if**

**Synchronized Network State Update:**
    Upon intrusion alert reception, Controllers synchronize their network states accordingly to achieve consistency.
    This synchronized response aids in prompt attack mitigation and network protection.

communication endeavours, mutual authentication among controllers is executed via their respective cryptographic keys to establish a foundation of trust. Subsequently, the "Communication Mode Selection" phase ensues. In this phase, the choice between two discrete communication modes is rendered: the General Communication Mode (GCM) and the Alert Communication Mode (ACM). While GCM is earmarked for routine network updates, ACM is specially tailored to accommodate exigent alerts associated with intrusion events.

The ensuing phase, denominated "Key Exchange and Synchronization," centres on the exchange of cryptographic keys during initialization. This process enables secure message encryption and decryption. Furthermore, a robust synchronization mechanism is enacted to ensure controllers possess the most current cryptographic keys, thereby engendering a state of secure and synchronized communication. The subsequent phase, designated as the "Decentralized Warning Conduit (DWC) Establishment," pertains to the establishment of the DWC within the ACM paradigm. Through secure Conduits rooted in the Advanced Message Queuing Protocol (AMQP), controllers concretize the DWC as a dedicated conduit for the transmission of intrusion alerts and critical network events. The "Intrusion Alert Propagation" phase follows suit. When a controller detects an intrusion attack, it encrypts an alert message utilizing the key associated with the DWC. This encrypted alert is then disseminated through the DWC, thereby apprising other participating controllers of the detected intrusion event.

The final phase, termed "Synchronized Network State Update," encompasses the network-wide synchronization that ensues upon receipt of an intrusion alert by other controllers. This synchronization underpins a swift and coordinated response to the detected intrusion, thus reinforcing the network's defence mechanisms. In conclusion, the proposed secure decentralized communication scheme (SDCS) tailored for Controller-to-Controller (C2C) communications within SD-IoT networks constitutes a transformative approach. By substantially curtailing communication overhead while concurrently bolstering the capabilities for intrusion attack identification and response, the integration of cryptographic techniques, decentralized architectural paradigms, and synchronized communication Conduits collectively fosters efficient and secure information exchange amongst participating controllers. Fig. 3 illustrates the execution logic of the proposed communication strategy, while Fig. 4 presents a visual roadmap illustrating the comprehensive flowchart of the MULTI-BLOCK framework.

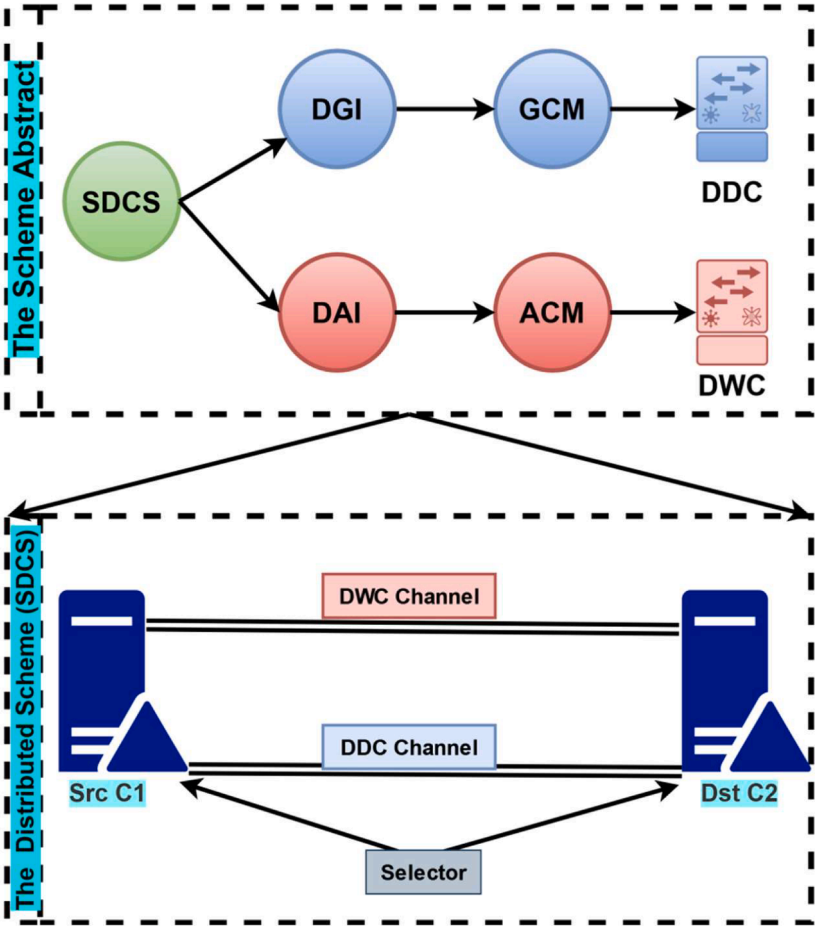



**Fig. 3.** Implementation of the proposed decentralized communication scheme (SDCS).

### 3.2. The second module (P4-DTMS)

The second module, Proposed P4-Enabled Decentralized Traffic Monitoring System (P4-DTMS), is designed to offer an advanced and efficient approach to managing IoT network traffic. The P4-DTMS module introduces an innovative Sequential Multi-State Pipeline (SMSP), serving as the central component of the P4-DTMS module. The SMSP consists of 24 P4-enabled state tables, each meticulously crafted to capture specific aspects of the network's traffic. These state tables collectively contribute to creating a comprehensive network overview that accounts for various features and attributes of the data flowing through the IoT network. These tables are thoughtfully designed to accommodate different types of traffic-related data, enabling the module to generate a holistic picture of the network. This module plays a pivotal role in gathering, processing, and providing vital insights into the network's traffic patterns. The P4-DTMS module algorithm listed in Algorithm 2 commences with the initialization phase, where the P4-DTMS module is established. The algorithm then delves into the state table configuration, which involves fine-tuning each state table to serve its designated purpose. This includes allocating memory and resources to store the collected information effectively. Subsequently, the traffic feature extraction process begins. Traffic data from diverse sources within the IoT network is collected, and pertinent features are extracted. These features encompass a range of details such as packet sizes, source and destination addresses, protocol types, and flow durations. Each extracted feature is then populated within the corresponding state tables. Fig. 5 presents a graphical representation of the P4-DTMS module implementation.

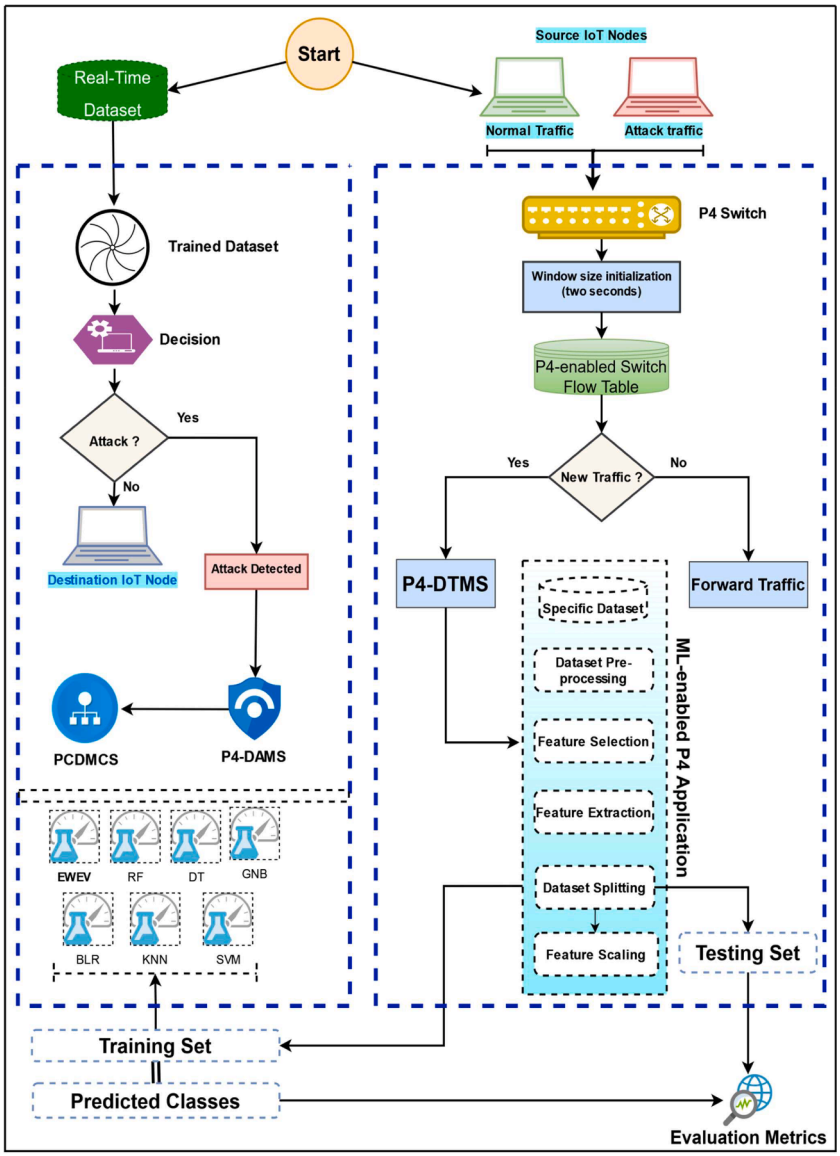



**Fig. 4.** A visual roadmap illustrating the comprehensive flowchart of the MULTI-BLOCK framework.

The algorithm's focal point lies in creating a comprehensive network overview by aggregating data across all 24 state tables. This aggregated information paints a detailed picture of the network's traffic behaviors, enabling administrators to discern patterns and trends. This comprehensive view is invaluable for effective network management and optimizing operational strategies. Furthermore, the algorithm elucidates the significance of the P4-DTMS module as a valuable resource for subsequent modules. It outlines how the synthesized network overview becomes a critical input for both the intrusion detection and traffic mitigation components. By channelling this overview into the intrusion detection system, potential anomalies and suspicious activities can be promptly identified and addressed.

Simultaneously, sharing this network-wide insight with other modules enhances its ability to distribute traffic optimally across the network and ensure efficient load balancing. In essence, the proposed algorithm for the P4-DTMS in SD-IoT underscores the essential role played by this module in orchestrating effective network management. Through meticulous state table configuration, data extraction, and comprehensive network overview creation, the P4-DTMS module becomes a cornerstone for subsequent actions, contributing significantly to the enhancement of network security and operational efficiency within the broader framework.

Table 2 presents the set of features that have been carefully selected to aid in intrusion detection within IoT networks, alongside their extraction/computation method. These features encompass a range of network-related characteristics and behaviors, all of which contribute to effectively identifying and mitigating potential intrusion activities. These features are divided into extracted and calculated types. Extracted features are directly derived from packet-level information or protocol attributes. For instance, features such as "RTT" (Round-Trip Time) and "TTL" (Time to Live) provide insights into network latency and packet routing, respectively. Features like "Protocol" and "PingStats" (Ping Statistics) offer information about the type of communication and network performance metrics. The calculated features involve complex calculations based on multiple packet-level attributes. Features such as "PktSzDist"

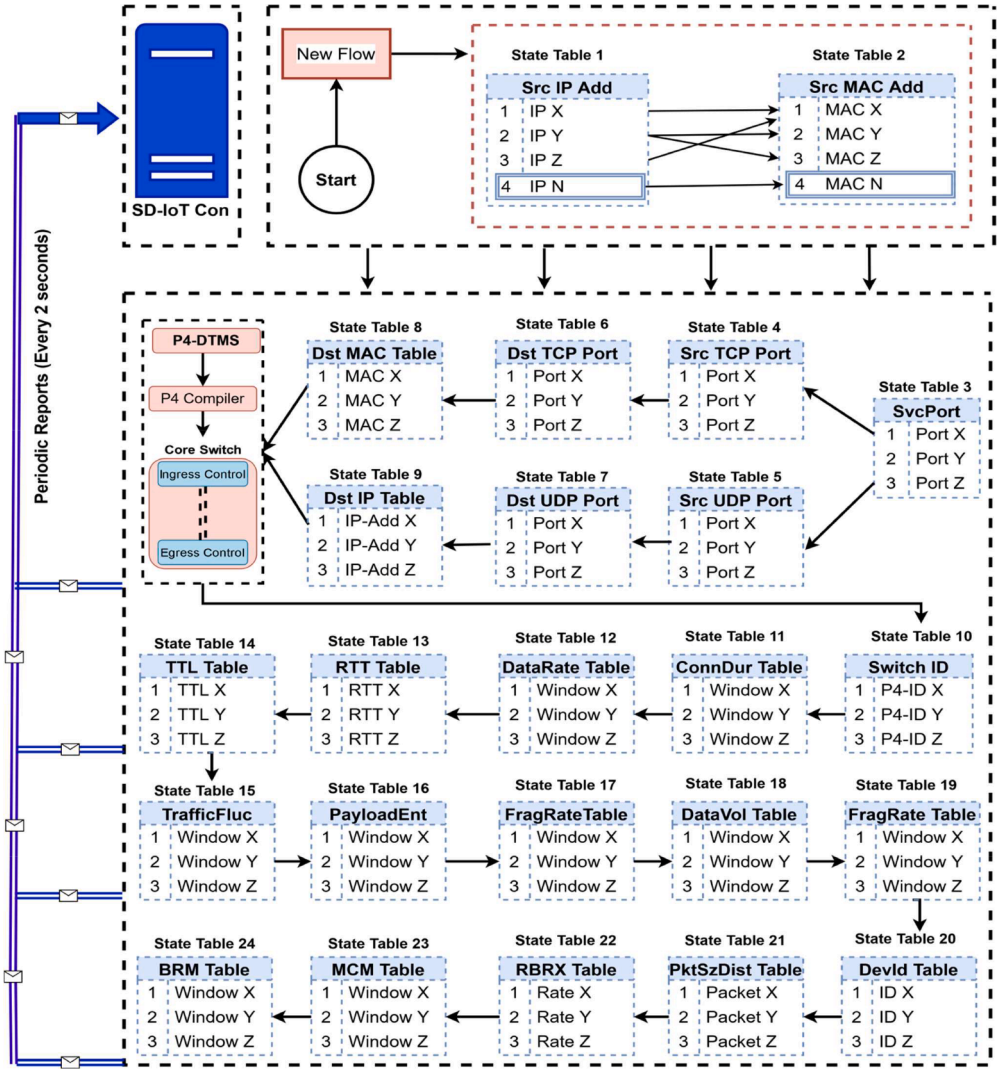



**Fig. 5.** Visualizing innovation - graphical representation of the P4-DTMS module implementation.

**Table 2**
The newly extracted and calculated feature set.

| No | F_Name | F_Description | F_Type | Methodology for feature extraction/calculation |
|---|---|---|---|---|
| 1 | MPCFW | Mean Packet Count per Flow Window | Calculated | MPCFW = (The ratio of packets across all flows within a window to the overall number of flows) |
| 2 | MBCFW | Mean Bytes Count per Flow Window | Calculated | MBCFW = (The ratio of bytes across all flows within a window to the overall number of flows) |
| 3 | PIRW | Rate of Packet-In instances per window | Calculated | PIRW = (Packet_In instances across all flows divided by the window time) |
| 4 | RBRX | Rx Bytes Rate | Extracted | RBRX = (Total number of received bytes per window time) |
| 5 | FERW | Rate of Flow Entry occurrences per window | Calculated | FERW = (The total count of flow entries divided by the duration of the time window) |
| 6 | DIPEW | Entropy of Dest_IP Addresses per window | Calculated | DIPEW = (Determined through the DIPEW function within the P4 code) |
| 7 | STPEW | Src TCP Port Entropy per window | Calculated | STPEW = (Calculated using STPEW function in P4 application) |
| 8 | SUPEW | Src UDP Port Entropy per window | Calculated | SUPEW = (Calculated using SUPEW function in P4 application) |
| 9 | HTTPPR | HTTP Packets Rate | Extracted | HTTPPR = (Total number of HTTP packets per time window) |
| 10 | TCPPP | TCP Packets Percentage | Calculated | TCPPP = (The ratio of the total count of TCP packets to the overall packets within a given window) |
| 11 | ICMPPP | ICMP Packets Percentage | Calculated | ICMPPP = (The ratio of the total count of ICMP packets to the overall packets within a given window) |
| 12 | HTTPPP | HTTP Packets Percentage | Calculated | HTTPPP = (The ratio of the total count of HTTP packets to the overall packets within a given window) |
| 13 | BRM | Broadcast Rate Metric | Extracted | BRM = (Total number of broadcast packets per time window) |
| 14 | MCM | Multicast Rate Metric | Extracted | MCM = (Total number of multicast packets per time window) |
| 15 | UPPM | Unicast to P4-Based Packet Ratio | Calculated | UPPM = (The ratio of unicast packets to the total number of P4-based packets) |
| 16 | ConnDur | Connection Duration | Calculated | Duration of the communication session between devices |
| 17 | DataRate | Data Rate | Calculated | Rate of data transmission within a flow |
| 18 | RTT | Round-Trip Time | Extracted | RTT = (average round-trip time between communicating devices) |
| 19 | TTL | Time to Live | Extracted | TTL = (number of hops a packet can travel before being discarded) |
| 20 | Protocol | Protocol Code | Extracted | Identification of the specific protocol code of the packet |
| 21 | PktSzDist | Packet Size Distribution | Calculated | Distribution of packet sizes within a flow |
| 22 | HdrSzDist | Header Size Distribution | Calculated | Distribution of header sizes within a flow |
| 23 | ReqRespRat | Request-Response Ratio | Calculated | Ratio of request packets to response packets in a flow |
| 24 | PingStats | Ping Statistics | Extracted | Metrics derived from ping requests, including round-trip time, packet loss. |
| 25 | DataVol | Data Volume | Extracted | Volume of data exchanged within each flow |
| 26 | FragRate | Fragmentation Rate | Extracted | Rate at which packets are fragmented during transmission |
| 27 | PayloadSz | Payload Size | Extracted | Size of the payload in each packet |
| 28 | Ipt | Inter-Packet Time | Extracted | Time interval between consecutive packets in a flow |
| 29 | DevId | Device Identification | Extracted | Unique identifiers of IoT devices involved in the communication |
| 30 | TrafficDir | Traffic Direction | Extracted | Direction of traffic flow (ingress or egress) |

(Packet Size Distribution) and "HdrSzDist" (Header Size Distribution) provide insights into packet size patterns, while "ReqRespRat" (Request-Response Ratio) and "PayloadEnt" (Entropy of Payload) aid in detecting abnormal communication patterns and payload characteristics.

Additionally, these features address the distinct challenges of IoT networks, where a vast number of heterogeneous devices communicate and interact. The "Protocol Code" feature aids in identifying unauthorized or unusual protocols, a critical aspect given the diversity of IoT device communication. The "Ping Statistics" feature provides deeper visibility into network performance, allowing for the timely detection of irregularities. The inclusion of features such as "Data Volume," "Fragmentation Rate," and "Payload Size" is particularly relevant in the IoT context, as these features help identify abnormal data transfer patterns and potential data fragmentation attacks that can target IoT devices. "Inter-Packet Time" assists in identifying sudden changes in communication patterns, which may signify an ongoing attack. "Device Identification" and "Traffic Direction" contribute to enhancing the system's capability to distinguish legitimate from suspicious device behaviors. Moreover, "Packet Size Distribution" and "Header Size Distribution" capture fine-grained details about packet and header variations, aiding in the identification of unusual traffic patterns indicative of intrusion attempts. "Request-Response Ratio" and "Connection Duration" enable the framework to detect abnormal request-response behaviors and prolonged communication sessions that may indicate malicious intent.

In conclusion, the adoption of these IoT-specific features significantly enriches the intrusion detection framework's ability to identify and mitigate potential threats within SD-IoT networks accurately. The comprehensive set of attributes provides a holistic view of network behaviors and anomalies, allowing the system to respond promptly to emerging security challenges. These features collectively empower the intrusion detection system to effectively safeguard the integrity, availability, and confidentiality of IoT networks in the face of evolving intrusion attempts. The data collected from these features are processed and scrutinized to assess whether they signify legitimate traffic or potential intrusion endeavours. The P4-enabled feature extraction techniques ensure that the selected attributes are efficiently captured, contributing to the success of the intrusion detection system within IoT.

### 3.3. The third module (P4-DIDPS)

This module uses thirty P4-extracted and calculated features to implement a unique machine learning (ML)-based state-driven P4 program for attack detection and prevention. The module employs an Enhanced Weighted Ensemble Algorithm (EWEA) to detect anomalies in the extracted traffic patterns. State-driven P4 enables accurate and timely identification of attacks at the data plane stage by providing the application with a comprehensive picture of the network state. The classification models and module stages used are discussed in the following subsections.

#### 3.3.1. P4-driven feature extraction stage

In machine learning, the feature extraction step is critical because it helps choose pertinent information from raw data so that the model can be trained successfully. We used the P4-driven feature extraction stage in the proposed framework to extract important characteristics from the network data. Specifically, our method yielded a collection of 30 unique features, consisting of 16 calculated and 14 extracted features. The feature extraction stage is a critical component that helps our proposed system to detect possible threats and then mitigate them.

When IoT nodes send data across the network, the traffic workflow begins. This traffic data is received and sent by P4 switches to the feature extraction stage. The module, in this case, carefully collects the necessary features and sends them to the EWEA classifier so that the traffic may be classified. The classification decision is then relayed back to the P4 switch so that it may take the necessary action, depending on whether the traffic is considered normal or suspicious of an attack.

The P4 code's procedural workflow proceeds as follows: The inbound packets are first parsed to retrieve relevant protocol headers. The information that has been processed makes its way into the ingress pipeline, which is comprised of 24 unique state tables, as specified in the P4-DTMS module. The purpose of each of these tables is to update the packet information and feature occurrences in registers. The ingress pipeline is conditionally evaluated when statistics are updated.

If the designated window closes, the code executes commands that result in the creation of a report that includes packet analysis and categorization. After then, this information is sent to the P4 switch. On the other hand, if the window period is still open, the process follows ordinary protocols, forwarding or stopping packets in accordance with the flow entries that the EWEA classifier has set.

Because of this orchestration, the P4 switch may act as a firewall in real time, deciding whether to allow or block suspicious flows based on the EWEA classifier module's conclusions. Specifically, our architecture makes use of the P4 switches to produce, handle, and store stateful features, which guarantees the system's coherent scalability. P4 switches are scalable and able to handle an increasing quantity of flow entries. This feature allows a large number of flows to be analyzed using the same control pipeline segment. Fig. 6 shows a visual depiction of the P4 program pipeline workflow with the feature extraction stage included.

#### 3.3.2. Training, testing, and validation stages

The Enhanced Weighted Ensemble Algorithm (EWEA) relies on a two-phase training and testing methodology to evaluate its intrusion detection capabilities. During the training phase, the preprocessed and labelled features extracted from IoT network traffic are divided into training and validation sets. Subsequently, various machine learning classifiers are trained on the training set. The validation set is employed to assess the performance of each classifier. This evaluation allows for the comparison of different classifiers and the selection of the most effective ones. A scoring mechanism, ranging from 0 to 1, is implemented to quantify the performance of each classifier on the validation data. This scoring system facilitates the determination of relative importance among the classifiers.

The testing phase leverages the trained classifiers to predict attack types within the testing set. This enables the assessment of the generalizability of the learned models on unseen data. A confusion matrix is constructed to evaluate the accuracy of the prediction.

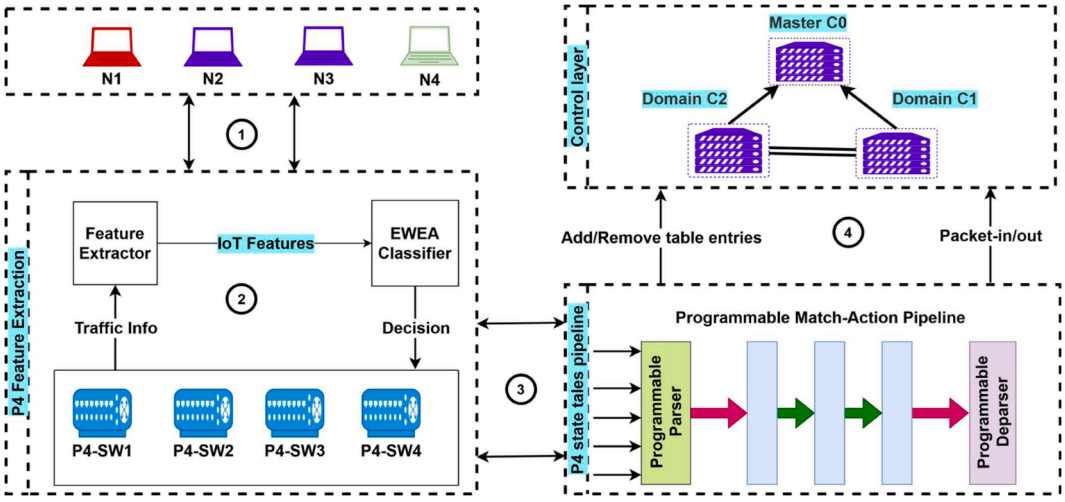



**Fig. 6.** Navigating the P4 pipeline - illustrating the workflow of the P4 program.

**Algorithm 2**

**Algorithm 3: The Intrusion Detection Algorithm (EWEA)**

**01 Inputs:**

02 Labeled IoT Data: Labeled dataset containing IoT network traffic data with corresponding class labels (attack or normal).
03 Incoming IoT Network Data: Real-time IoT network traffic data for intrusion detection.
04 Threshold: Threshold value for class label prediction.

**05 Output:**

06 Intrusion Detection Results: Classification of incoming IoT network data as attack or normal based on the algorithm's decision.
07 Performance metrics for assessing algorithm effectiveness.

08 IoT_S ← IoT Selected Features
09 IoT_T ← IoT Total traffic features
10 SF_T ← IoT Selected Features Table
11 IoT_Labeled ← IoT Labeled Traffic
12 WC_Label ← Weighted Class Label

**13 Initialization:**

14 Define the set of ML classifiers.
15 Set an initial threshold for class label prediction.

**16 Feature Selection and Extraction:**

17 **For each** (IoT_T):
18 **Preprocess** IoT network traffic data by removing irrelevant and redundant features.
19 **Employ** Recursive Feature Elimination to retain pertinent features.
20 **Select** IoT_S from IoT_T
21 **Keep** IoT_SF in SF_T
22 **End for**

**23 The Training Stage:**

**24 For each** (IoT_Labeled):
25 **Divide** the labelled dataset into validation and training subsets.
26 **Train** the classifiers on the training subset using selected features.
27 **Evaluate** the performance of each classifier on the validation subset using appropriate metrics.
28 **Calculate** the weights for each classifier based on its performance metrics.
29 **Normalize** the classifier weights to ensure a cumulative value of 1.
30 **End for**

**31 The Detection Stage:**

32 **Receive** incoming IoT network traffic features via P4.
33 **Extract** relevant features from the received data.
34 **For each (Classifier)**
35 **Make use** of the trained model to forecast the extracted features' class labels**.**
36 **Multiply** the weight assigned to the projected class label**.**
37 **Calculate** the weighted cumulative class labels.
**38 End for**

**39 The Decision-Making Stage:**

40 **For each (WC_Label)**
41 **To arrive** at a final estimated class label, add the weighted class labels.
42 **Compare** the predefined threshold with the final estimated class label.
43 **If ( Final label > threshold)**
44 **Classify** the traffic as an attack.
45 **Else**
46 **Classify** the traffic as normal.
47 **End for**

**48 Evaluation Stage:**

49 **Evaluate** the algorithm's performance using a separate test dataset.
50 **Employ** different evaluation criteria.

**51 Result Analysis Stage:**

52 **Analyze** the experimental results to assess the algorithm's effectiveness.
53 **Compare** the proposed EWEA algorithm's performance against conventional ensemble voting methods.
54 **Measure** the improvement in detection accuracy and overall efficacy in SD-IoT network intrusion detection.

**55 Conclusion:**

56 **Deduce** from the experimental data that the EWEA algorithm improves the accuracy of intrusion detection**.**
57 **Emphasize** its superiority over traditional ensemble voting methods in terms of accuracy and performance.

This matrix visualizes the distribution of correct and incorrect predictions across different attack categories, providing insights into the strengths and weaknesses of the classification models. The trained models persist in using *joblib*, a utility from the *scikit-learn* library. This process preserves the knowledge acquired during training, allowing for future deployment of the EWEA algorithm for real-world intrusion detection. The entire training and testing methodology facilitates a comprehensive evaluation of the EWEA's effectiveness in identifying and classifying diverse intrusion attacks within IoT networks.

For the validation, we employed k-fold cross-validation to ensure the robustness and generalizability of our results. In k-fold cross-validation, the dataset is randomly shuffled and divided into k-equal folds. The model is then trained on k-1 folds and evaluated on the remaining fold. This process is repeated $k$ times, with each fold serving as the testing set exactly once. To determine the optimal number of folds ($k$), we experimented with values ranging from 5 to 20. We chose $k = 10$, which provides a good balance between bias and variance. We utilized a well-established benchmark dataset for IoT intrusion detection, such as Edge-IIoTset. This dataset provides a comprehensive set of labelled network traffic samples. We divided the dataset into training and testing sets using an 80/20 split (80 % for training and 20 % for testing). To assess the effectiveness of the proposed method, we employed standard metrics like accuracy, precision, recall, and F1-score.

#### *3.3.3. Intrusion detection stage*

A new method called the Enhanced Weighted Ensemble Algorithm (EWEA) is presented to identify intrusions in SD-IoT networks. By combining predictions from many machine learning classifiers, such as DT, SVM, BLR, kNN, RF, and GNB, this technique seeks to improve detection accuracy. This joint effort improves the total precision of detection. The algorithm is notably customized with the use of a dynamic weighted voting mechanism (DWVM). Based on how well each classifier performs on the training dataset, their weights are adaptively changed.

Feature selection is the initial stage of the EWEA algorithm, a strategic move that reduces computing complexity while increasing detection efficacy. Information gleaned from IoT network traffic features necessitates the removal of redundant and unnecessary attributes. Following this pruning process, relevant features are selected using Recursive Feature Elimination (RFE), a feature selection method deemed appropriate for intrusion detection in IoT for several reasons. Firstly, RFE facilitates dimensionality reduction by iteratively eliminating the least significant features, thereby reducing the dataset's dimensionality. In IoT environments, where resources may be limited, minimizing the number of features is crucial for efficient processing. Secondly, RFE seamlessly integrates with various machine learning classifiers, enhancing the interpretability of the model by focusing on a reduced feature set. This makes it easier to comprehend and analyze the model in the context of intrusion detection. Thirdly, RFE is adaptable to different machine learning algorithms, making it suitable for diverse intrusion detection scenarios in IoT where data characteristics may vary. Fourthly, by selecting the most relevant features, RFE can improve the performance of machine learning models by concentrating on key aspects of the data related to intrusion patterns. Lastly, in resource-constrained IoT environments, RFE contributes to creating more resource-efficient models by selecting features that significantly contribute to the detection task.

The labelled IoT network traffic characteristics are then divided into validation and training subsets during the training phase. The chosen training dataset with the chosen characteristics trains each machine learning classifier. The performance of the classifiers is then evaluated using the validation set. Each classifier's weight is established by considering its performance measures; these weights are then standardized. During the detection stage, P4 is used to extract pertinent characteristics from incoming IoT network data after features are gathered. In this case, the purpose of each ML classifier is to forecast the class labels for the supplied traffic characteristics. After that, these forecasts are multiplied by the appropriate weights. A final projected class label is produced by adding all the weighted class labels. After that, this label is evaluated in comparison to a preset threshold. The traffic is classified as an attack if the anticipated class label is greater than the threshold.

On the other hand, the traffic is shown as usual if it drops below. The evaluation stage represents the algorithm's apex. The performance of the proposed approach is examined using a test dataset. A variety of assessment criteria are used to assess its performance, which will be further discussed in the upcoming section on the experimental results. These experimentally-derived data confirm how the proposed EWEA method greatly improves detection accuracy. Interestingly, it performs better than the traditional ensemble voting method when it comes to SD-IoT network intrusion detection. The EWEA algorithm's successive stages are outlined in detail in Algorithm 3, which is shown below Algorithm 2.

### *3.4. The fourth module (P4-DAMS)*

An SD-IoT network's decentralized mitigation mechanism for thwarting hostile traffic flows is outlined in the P4-enabled decentralized attack mitigation strategy (P4-DAMS) algorithm. Following the detection of such traffic flows, a sequence of activities commences, including warning propagation, flow table adjustments, isolation of compromised nodes, and communication blockage.

By utilizing the MULTI-BLOCK architecture for decentralized administration and dynamic modifications, the offered mitigation algorithm emphasizes the importance of strong and efficient IoT network security. After recognizing a malicious traffic flow, the detection module notifies the mitigation module, which then acts upon the information. Its goal is to protect IoT nodes by adding more high-priority flow rules to the P4-switch flow tables that correspond to the attack packets' rules. The switch blocks all packets coming from the attack source by creating a new entry in the flow table in response to the controller sending it a FlowMod message. The consequences of the strike are lessened and lessened by this action. This module also isolates the compromised device by rerouting traffic and cutting it off from the network. DWC enters the picture during process execution, sending alert signals to other controllers. Furthermore, by employing the pyramidal MULTI-BLOCK design, the module's capabilities are expanded to include a wider range of controllers, enabling the system to handle large-scale attacks and guarantee prompt replies effectively.

Simultaneously, Algorithm 4 outlines the steps that the attack mitigation mechanism goes through. First, the procedure uses the information from the cooperation between P4 switches and controllers to classify traffic into three different categories: attack traffic, regular traffic, and new traffic. The system examines each new entry of IoT traffic to see if it fits into the trends of attack traffic. If a match is found, the algorithm will isolate the affected IoT nodes, reroute traffic, implement new flow rules, remove packets that originate from the attack source, block malicious source MAC, source IP, and source port connections, and send out alerts. From a holistic perspective, the mitigation algorithm offers a structured strategy for the detection and counteraction of malevolent flows. The integration of P4-enabled switch state tables, the DWC, and the system's decentralized architecture collectively bolster effective attack mitigation and elevate the security posture of the network Algorithm 3.

**Algorithm 3**

**Algorithm 4: The P4-DAMS Module Algorithm**

**Inputs:** DWC, P4-IoT-NT, P4-IoT-WT, P4-IoT-FT, and P4-IoT-CT

**Output:** Mitigated Attack IoT Network Traffic

**Initialization:**

**Obtain** P4-IoT-NT (P4-Enabled SD-IoT Network Traffic).
**Acquire** P4-IoT-FT (P4-Enabled SD-IoT Flow Tables).
**Access** P4-IoT-CT (P4-Enabled SD-IoT Controllers).
**Fetch** P4-IoT-WT (P4-Enabled SD-IoT Time Window).
**Utilize** DWC (Decentralized Warning Conduit).
**Deploy** P4-DAMS (P4-Decentralized Attack Mitigation Strategy).

**Detect Malicious Traffic Flow:**

**For each (Traffic_Entry) in P4-IoT-NT:**
    **If (Traffic_Entry == malicious patterns):**
        **Proceed** to the next step (Activate Mitigation Process).
    **End if**
**End for**

**Activate Mitigation Process:**

**For each (Malicious Traffic_Entry):**
    **Isolate** the affected IoT nodes.
    **Redirect** network traffic to ensure isolation.
    **Utilize** DWC to broadcast alert notifications to controllers.
**End for**

**Implement Dynamic Flow Table Modification:**

**For each (Malicious Traffic_Entry):**
    **Identify** the attack packet rules.
    **Modify** the corresponding P4-IoT-FT with high-priority rules.
    **Update** P4-IoT-FT by adding new flow table entries.
    **Drop** packets originating from the source of the attack.
**End for**

**Enhance Security Measures:**

**For each (Malicious Traffic_Entry):**
    **Block** communication of source MAC, source IP, and source port.
    **Publish** warning alert through DWC to notify other controllers.
**End for**

**Distribute Mitigation Across Controllers:**

**Employ** MULTI-BLOCK to distribute the mitigation process.
**Enable** controllers to handle large-scale attacks collaboratively.

**Evaluate and Optimize:**

**Continuously** monitor the effectiveness of the mitigation process.
**Adaptively** adjust mitigation strategies as needed.

**Conclude:**

**Mitigate** malicious traffic flows successfully.
**Enhance** network security and resilience.

## 4. Performance results and evaluation

### 4.1. Experimental settings

Using an Intel Core i7-1355U CPU and 12 GB of memory, a virtual machine operating Ubuntu 20.04.6 LTS was used for the experiment. Software tools used to build and run the IoT testing scenarios included sFlow-RT, Mininet-Wifi, Wireshark, P4 compiler, and VMware Workstation 17 Pro. Table 3 provides the modules' experimental configurations inside the proposed framework. Moreover, Fig. 7 offers a schematic of the p4-BLOCK framework's SDN-enabled IoT topology. To fully evaluate the effectiveness of the suggested detection framework in this context, four different test scenarios were conducted: 1) Single-Host Victim Attack (SHVA), 2) Two-Host Victim Attack (THVA), 3) Four-Host Victim Attack (FHVA), and 4) Eight-Host Victim Attack (EHVA).

### 4.2. The experimental test cases

#### 4.2.1. The single-host victim attack (SHVA) test case

In the SHVA test case, we delve into a specific intrusion scenario where a single IoT node is subjected to a targeted attack. The primary objective of this test case is to evaluate the framework's efficiency in promptly detecting and mitigating intrusion attempts directed at a singular IoT device while capitalizing on the collaborative nature of the multi-controller architecture. Our experimental setup emulated a sophisticated IoT topology comprising five Ryu controllers. Additionally, four access points (APs), two P4-enabled switches, and a network of 40 IoT nodes were strategically deployed. This network architecture facilitated a comprehensive assessment of the proposed detection framework's capabilities through diverse test scenarios. To initiate the SHVA scenario, we orchestrated a deliberate intrusion on a designated IoT node, seeking to compromise its communication or inflict disruption. Operating within the specified IoT network topology, we established a specific attack layout using IoT Node 1 (N1) as the attacker and IoT Node 5 (N5) as the intended target. The communication Conduits of Nodes N6, N8, and N10 were designated as authorized pathways for normal traffic circulation. The P4 program is run via the "P4-1-enabled" switch, as seen in Fig. 7. To evaluate the framework's performance, we launched various attacks on Node N5. These attacks included DDoS attacks, spoofing attacks, and intrusions orchestrated by a botnet. We used the pingall command to generate ICMP packets, simulating normal network traffic. Real-time P4 features were then collected for analysis. To simulate actual intrusion attempts, we employed the hping3 tool. For the DDoS attack, we flooded the network with traffic at a rate of 20 % of the total flow directed towards Node N5.

Our assessment encompassed various traffic features, including DIPEW, STPEW, and SUPEW, which respond to entropy dynamics. During intrusion attempts, certain features exhibited alterations, including decreases or increases, signifying deviations from the normal network behavior. We simulated a Botnet attack, where compromised IoT devices were controlled remotely to launch malicious activities like spreading spam or participating in DDoS attacks. We closely monitored the effects of these attacks, focusing on unusual device behavior, atypical traffic patterns, and increased traffic directed toward command and control servers.

Fig. 8 visually highlights the impact of the Botnet scenario (SHVA). The Mininet dashboard displayed a regular traffic volume of around 150 kbps. However, during the attack, traffic surged significantly to over 40 Mbps. This demonstrates the framework's effectiveness in identifying and responding to targeted intrusion attempts. The SHVA test case evaluates a single compromised IoT node within a multi-controller architecture. By doing so, it helps assess the framework's effectiveness in protecting against targeted attacks.

#### 4.2.2. The two-host victim attack (THVA) test case

The THVA test case introduces a scenario wherein an attacker simultaneously targets two distinct IoT nodes. This scenario is

**Table 3**
Configuration parameters - experimental settings for the modules within the proposed framework.

| Experiment setting | Features | Used module |
|---|---|---|
| Topology size | 40 IoT nodes | All Modules |
| Virtual Machine | VM Ware Workstation 17 Pro | All Modules |
| Operating System | Ubuntu 20.04 LTS | All Modules |
| CPU | Intel Core i7-1355U Processor | All Modules |
| RAM | 8 GB | All Modules |
| Wireshark 2.2.0 | To capture the packets | P4-DTMS and P4-DIDPS |
| Xterm | IoT node interface emulator | P4-DIDPS |
| Bmv2 | P4 software switch | All Modules |
| Scapy | For generating attack scripts | P4-DIDPS |
| hping3 | Used for attack traffic generation | P4-DIDPS |
| P4 language | 16 version | All Modules |
| Mininet-Wifi 2.2.0 | SD-IoT network emulator | All Modules |
| Python 3.7 | To build complex configuration | All Modules |
| sFlow-RT | For traffic monitoring | P4-DTMS |
| Dataplane switch | Six P4-Switches | All Modules |
| SD-IoT Controller | Five Controllers, One Master, Two Domains, and Two Backup. | All Modules |
| Access Points | Four Access Points | All Modules |

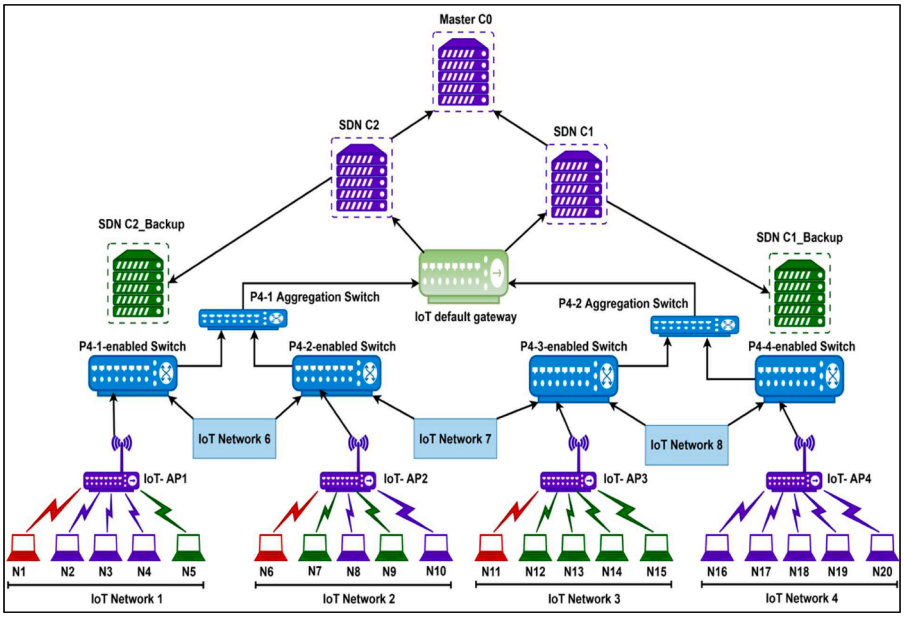



**Fig. 7.** Illustrating the blueprint of SDN-enabled IoT topology in the P4-BLOCK framework.

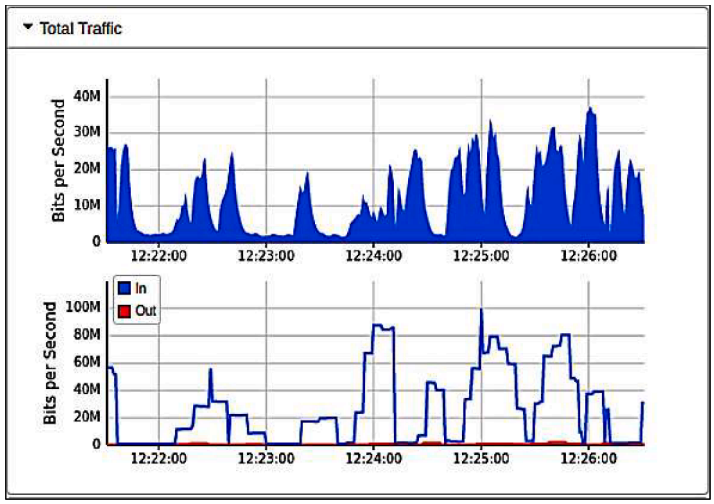



**Fig. 8.** Monitoring the storm - mininet dashboard showcasing overall traffic amidst attack operating conditions.

designed to assess the framework's ability to handle coordinated attacks and accurately identify multiple intrusion points concurrently. In the THVA test case, our topology stays in line with the earlier configurations. However, the crucial distinction lies in the coordinated attack on two specific IoT nodes, N7 and N9, executed by N10. This intricate attack model simulates a real-world scenario where an attacker seeks to exploit vulnerabilities in multiple nodes simultaneously, potentially causing more widespread disruption or unauthorized access.

Moreover, the test scenario demonstrated the framework's efficacy in mitigating coordinated attacks. As the framework identified and responded to the intrusion attempts, the network traffic displayed distinct drops, indicating successful intervention. This dynamic response of network traffic further substantiates the effectiveness of deploying the P4-DIDPS component of the framework. In conclusion, the THVA test case served as a crucial evaluation of the framework's capabilities in managing simultaneous attacks on multiple targets. By successfully detecting, mitigating, and responding to such complex coordinated attacks, the framework proves its robustness and versatility in safeguarding SD-IoT networks against a diverse range of intrusion scenarios.

### *4.2.3. The FHVA and EHVA test cases*

In the FHVA scenario, the assailant directs their efforts toward compromising the security of four distinct IoT nodes that are

decentralized across the network. Conversely, in the EHVA scenario, the aggressor sets their sights on eight unique IoT nodes situated within the SDN-based IoT network. These attack instances simulate a range of potential assault methodologies intended to exploit vulnerabilities and disrupt the communication protocols of the targeted nodes. The objectives behind these orchestrated attacks encompass the intention to either overwhelm the network infrastructure with a barrage of malicious data or concurrently compromise multiple nodes. These test cases were meticulously designed to gauge the competence and effectiveness of the proposed framework for controlling and lessening the repercussions of multifaceted attacks. These attacks were simulated with varying intensity levels, spanning from 30 % to 90 % attack rates, mirroring real-world scenarios.

A spectrum of intrusion attack types was employed for these experiments, including but not limited to DDoS-based UDP attacks, DDoS-based TCP attacks, DDoS-based ICMP attacks, MiTM, and DDoS-based HTTP attacks. The deliberate omission of an attack rate of 20 % was justified by the fact that this distribution among four or eight victim nodes would result in a packet influx within the expected parameters of normal network behavior. It is noteworthy that as the simulation continued, the network's traffic requirements grew exponentially, leading to significant utilization of both P4 switches and controllers.

Fig. 10 showcases the negative impact of intrusion attacks on network operations, leading to resource saturation. Network measurements revealed complete CPU utilization, 63 % RAM consumption, a surge in bandwidth to 685 Kbps, and an extremely high transmission rate of 66 pps. However, Fig. 9 demonstrates the P4-DIDPS module, a core component of our framework, swiftly identifying the malicious activity and effectively restoring the network to its pre-attack state.

Furthermore, Fig. 11 illustrates the dynamic changes in traffic patterns during and after an attack. Noticeable spikes represent attack-generated traffic, while the following dips highlight the P4-DIDPS module's efficiency in mitigating the attack's impact. This functionality successfully detected and neutralized the attack, allowing the P4 switch to resume normal network operations. The results in Fig. 11 strongly support the P4-BLOCK framework's ability to promptly detect and thwart multi-target attacks, achieving restoration within a mere three milliseconds.

### 4.3. The P4-DIDPS module evaluation

In the proposed approach, fourteen distinct metrics were measured to assess the P4-DIDPS module's performance. The following describes these metrics, their definition, and their calculation process:

**Accuracy:** Accuracy reflects the proportion of correctly identified intrusion instances in the IoT network. It gauges the overall correctness of the intrusion detection model by considering both true positives and true negatives. The mathematical expression is shown in Eq. (1)

$$ACC = \frac{TP + TN}{FP + FN + TP + TN} \tag{1}$$

**Precision:** Precision signifies the accuracy of the model in identifying and flagging actual intrusions. It highlights the ratio of correctly identified intrusions to all instances flagged as intrusions. The mathematical expression is shown in Eq. (2)

$$PRE = \frac{TP}{FP + TP} \tag{2}$$

**Recall:** Recall represents the model's capability to identify and capture all actual intrusion instances correctly. It emphasizes the sensitivity of the model to true positive intrusions. The mathematical expression is shown in Eq. (3)

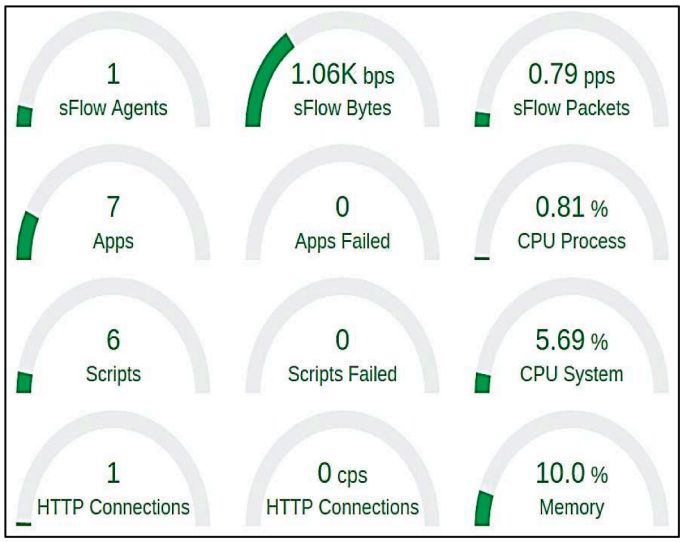



**Fig. 9.** Restoring order - IoT network resource status post-attack detection and mitigation.

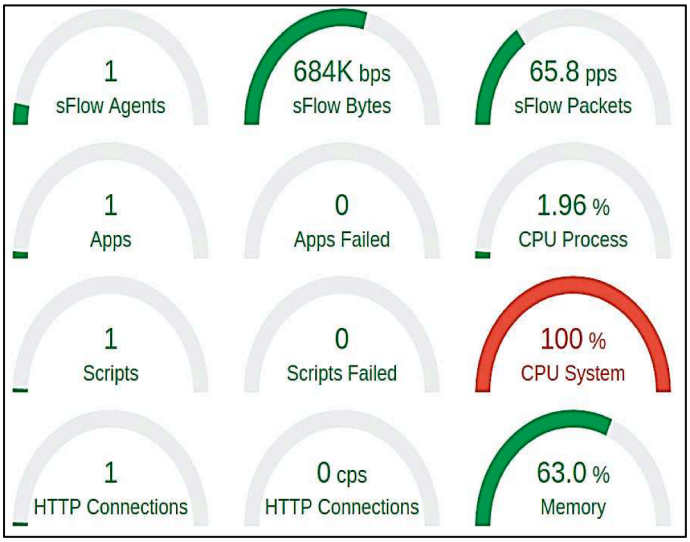



**Fig. 10.** Illuminating the impact - status of IoT network resources amidst attack conditions.

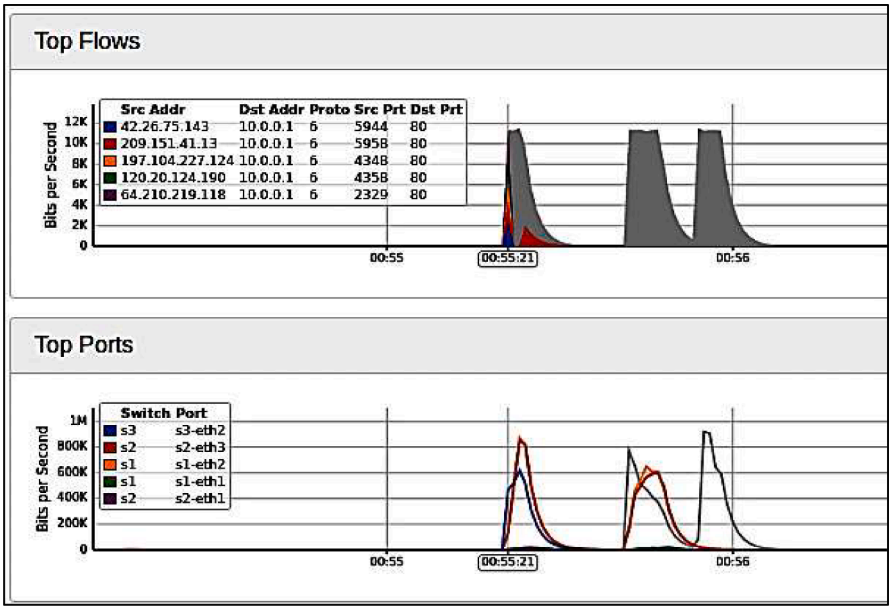



**Fig. 11.** Shield in action - visualizing the mitigation effect.

$$REC = \frac{TP}{FN + TP} \tag{3}$$

**F1-score:** The F1-score provides a balanced measure of the intrusion detection model's precision and recall. It offers insights into the model's ability to harmonize precision and capture intrusions effectively. The mathematical expression is shown in Eq. (4)

$$F1 = 2 * \frac{(PRE * REC)}{(PRE + REC)} \tag{4}$$

**True Negative Rate (TNR):** TNR is particularly significant in the IoT landscape, as it signifies the model's specificity in correctly identifying instances that are not intrusions, preventing false alarms in the network. The mathematical expression is shown in Eq. (5)

$$TNR = \frac{TN}{FP + TN} \tag{5}$$

**Negative Predictive Value (NPV):** NPV in IoT intrusion detection characterizes the model's ability to correctly identify instances

that are not intrusions among all instances predicted as non-intrusions. The mathematical expression is shown in Eq. (6)

$$NPV = \frac{TN}{FN + TN} \tag{6}$$

**False Positive Rate (FPR):** FPR indicates the percentage of legitimate network traffic misclassified as intrusion instances. It emphasizes the importance of minimizing false positives to avoid unnecessary alarms. The mathematical expression is shown in Eq. (7)

$$FPR = \frac{FP}{TN + FP} \tag{7}$$

**False Discovery Rate (FDR):** FDR reveals the proportion of instances misclassified as intrusions among all instances predicted as intrusions. It complements precision by highlighting the cost of false positives. The mathematical expression is shown in Eq. (8)

$$FDR = \frac{FP}{TP + FP} \tag{8}$$

**False Negative Rate (FNR):** FNR points out the ratio of instances incorrectly identified as non-intrusions among all actual intrusion instances. It underscores the model's ability to avoid missing actual intrusions. The mathematical expression is shown in Eq. (9)

$$FNR = \frac{FN}{TP + FN} \tag{9}$$

**Matthews Correlation Coefficient (MCC):** MCC offers a balanced measure of the intrusion detection model's performance, considering true positives, true negatives, false positives, and false negatives in the IoT context. The mathematical expression is shown in Eq. (10)

$$MCC = \frac{(TN * TP) - (FN * FP)}{\sqrt{((FN + TP) * (TP + FP) * (FP + TN) * (FN + TN))}} \tag{10}$$

**Average Detection Time (ADT):** ADT captures the average time taken by the intrusion detection system in IoT to identify and respond to intrusion instances, influencing the timeliness of the security framework. ADT was calculated using a specific function integrated into the P4 application. The mathematical expression is shown in Eq. (11)

$$ADT = \frac{\sum_{j=1}^{TET} SIDT}{TET} \tag{11}$$

**Attack Mitigation Effectiveness (AME)**: AME denotes the percentage of successfully mitigated intrusion instances in the IoT

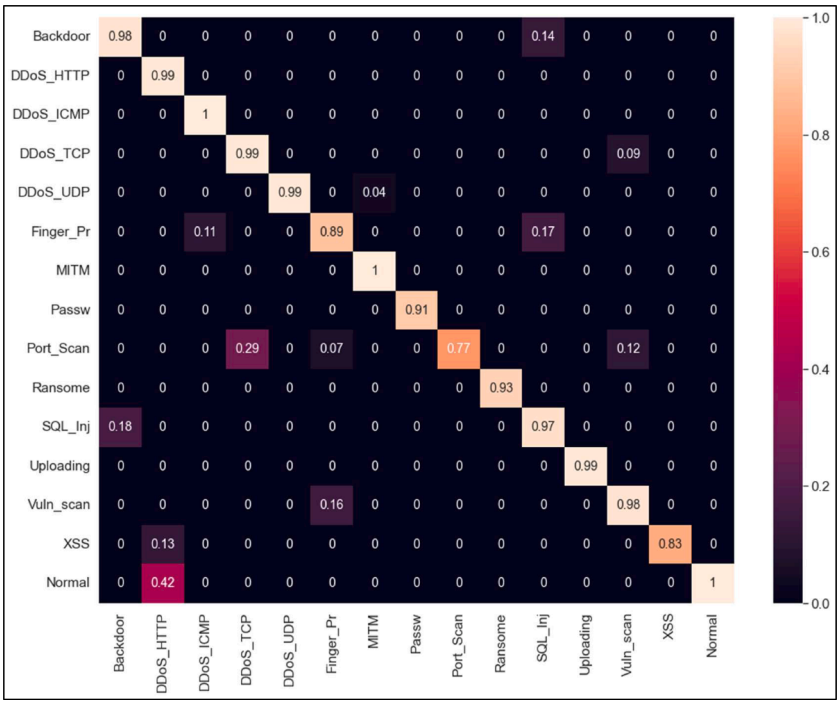



**Fig. 12.** The confusion matrix for the multi-class classification.

**Table 4**
Comparative evaluation results of seven ML classifiers in binary classification across three datasets.

| Dataset | Model | Accuracy | Precision | F1-score | Recall | Specificity | NPV | AUC | MCC | FPR | FDR | FNR | ADT(ms) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Edge-IIoTset | EWEA | 99.75 % | 99.32 % | 99.53 % | 99.67 % | 99.60 % | 99.95 % | 99.83 % | 99.14 % | 0.40 % | 0.68 % | 1.33 % | 2.11 |
| | RF | 98.86 % | 98.35 % | 98.60 % | 99.87 % | 98.55 % | 99.41 % | 99.59 % | 97.64 % | 3.45 % | 2.65 % | 3.53 % | 17.59 |
| | DT | 97.99 % | 97.21 % | 97.60 % | 97.63 % | 97.70 % | 97.48 % | 98.08 % | 96.45 % | 5.30 % | 2.79 % | 4.37 % | 27.09 |
| | KNN | 97.68 % | 96.83 % | 97.26 % | 97.10 % | 97.17 % | 96.93 % | 97.14 % | 95.60 % | 4.83 % | 4.17 % | 5.90 % | 18.95 |
| | GNB | 97.48 % | 96.39 % | 97.08 % | 94.71 % | 96.86 % | 96.65 % | 96.76 % | 94.92 % | 6.14 % | 5.61 % | 8.29 % | 12.67 |
| | SVM | 96.22 % | 95.04 % | 95.61 % | 95.45 % | 95.49 % | 95.34 % | 95.48 % | 92.45 % | 11.51 % | 7.96 % | 6.55 % | 55.72 |
| | BLR | 92.79 % | 93.87 % | 94.37 % | 95.51 % | 93.08 % | 93.05 % | 92.63 % | 88.29 % | 9.92 % | 11.13 % | 12.49 % | 21.04 |
| TON_IoT | EWEA | 98.17 % | 97.63 % | 97.90 % | 98.08 % | 98.11 % | 97.97 % | 98.03 % | 96.31 % | 1.89 % | 2.37 % | 1.92 % | 14.22 |
| | RF | 97.46 % | 96.80 % | 97.24 % | 97.06 % | 97.16 % | 96.75 % | 97.09 % | 94.73 % | 2.84 % | 3.20 % | 2.94 % | 39.11 |
| | DT | 96.65 % | 95.58 % | 96.14 % | 95.96 % | 96.02 % | 95.72 % | 95.89 % | 93.06 % | 3.98 % | 4.42 % | 4.04 % | 44.36 |
| | KNN | 95.51 % | 94.07 % | 94.87 % | 94.73 % | 94.80 % | 94.52 % | 94.67 % | 91.74 % | 5.20 % | 5.93 % | 5.27 % | 52.91 |
| | GNB | 95.15 % | 93.67 % | 94.60 % | 93.91 % | 94.21 % | 93.83 % | 94.01 % | 90.62 % | 5.79 % | 6.33 % | 6.09 % | 49.13 |
| | SVM | 94.01 % | 92.58 % | 93.27 % | 93.10 % | 93.17 % | 92.97 % | 93.07 % | 89.48 % | 6.83 % | 7.42 % | 6.90 % | 67.44 |
| | BLR | 88.62 % | 88.24 % | 89.90 % | 88.12 % | 88.22 % | 88.32 % | 88.17 % | 86.04 % | 11.78 % | 11.76 % | 11.88 % | 26.58 |
| X-IIoTID | EWEA | 98.21 % | 98.87 % | 97.54 % | 98.98 % | 97.12 % | 96.73 % | 97.68 % | 95.82 % | 5.88 % | 7.13 % | 6.02 % | 18.67 |
| | RF | 93.76 % | 92.35 % | 93.04 % | 93.26 % | 93.35 % | 91.87 % | 93.06 % | 90.98 % | 6.65 % | 7.65 % | 6.74 % | 31.12 |
| | DT | 92.84 % | 91.26 % | 92.01 % | 92.39 % | 92.48 % | 90.67 % | 91.98 % | 89.32 % | 7.52 % | 8.74 % | 7.61 % | 29.55 |
| | KNN | 91.59 % | 89.84 % | 90.68 % | 90.86 % | 90.93 % | 89.05 % | 90.45 % | 87.96 % | 9.07 % | 10.16 % | 9.14 % | 26.81 |
| | GNB | 90.72 % | 88.76 % | 89.72 % | 89.91 % | 89.97 % | 88.07 % | 89.37 % | 86.62 % | 10.03 % | 11.24 % | 10.09 % | 24.98 |
| | SVM | 89.34 % | 87.22 % | 88.25 % | 88.47 % | 88.53 % | 86.62 % | 88.01 % | 85.03 % | 11.47 % | 12.78 % | 11.53 % | 22.71 |
| | BLR | 87.68 % | 85.34 % | 86.55 % | 86.74 % | 86.80 % | 84.92 % | 86.03 % | 82.86 % | 13.20 % | 14.66 % | 13.26 % | 20.18 |

**Table 5**
Comparative evaluation results of seven ML classifiers in multi-class classification across edge-IIoTset dataset.

| Attack type | Model | Accuracy | Precision | F1-score | Recall | Specificity | NPV | AUC | MCC | FPR | FDR | FNR | ADT(ms) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| DDoS | EWEA | 98.72 % | 97.81 % | 98.26 % | 97.35 % | 99.05 % | 97.62 % | 98.01 % | 97.56 % | 0.95 % | 2.19 % | 2.65 % | 5.12 |
| | RF | 97.85 % | 96.47 % | 97.11 % | 96.83 % | 98.21 % | 96.62 % | 97.12 % | 96.74 % | 1.79 % | 3.53 % | 3.17 % | 8.35 |
| | DT | 96.91 % | 95.23 % | 96.02 % | 95.68 % | 97.12 % | 95.61 % | 96.02 % | 95.97 % | 2.88 % | 4.77 % | 4.32 % | 14.67 |
| | KNN | 96.15 % | 94.76 % | 95.45 % | 94.83 % | 96.47 % | 94.63 % | 95.78 % | 94.68 % | 3.53 % | 5.24 % | 5.17 % | 12.45 |
| | GNB | 95.36 % | 93.43 % | 94.36 % | 93.72 % | 95.48 % | 93.64 % | 94.23 % | 93.76 % | 4.52 % | 6.57 % | 6.28 % | 20.12 |
| | SVM | 94.48 % | 92.06 % | 93.26 % | 92.67 % | 94.71 % | 92.48 % | 93.45 % | 92.76 % | 5.29 % | 7.94 % | 7.33 % | 25.87 |
| | BLR | 91.97 % | 88.42 % | 90.10 % | 89.02 % | 92.39 % | 88.67 % | 89.85 % | 88.97 % | 7.61 % | 11.58 % | 10.98 % | 36.52 |
| Backdoor | EWEA | 98.15 % | 97.32 % | 97.68 % | 97.12 % | 98.34 % | 97.23 % | 97.85 % | 97.06 % | 1.66 % | 2.68 % | 2.88 % | 5.21 |
| | RF | 97.29 % | 96.14 % | 96.70 % | 96.28 % | 97.61 % | 96.13 % | 96.88 % | 96.07 % | 2.39 % | 3.86 % | 3.72 % | 8.97 |
| | DT | 96.42 % | 95.02 % | 95.72 % | 95.41 % | 96.83 % | 95.34 % | 95.96 % | 95.29 % | 3.17 % | 4.98 % | 4.59 % | 13.54 |
| | KNN | 95.68 % | 93.98 % | 94.78 % | 94.17 % | 96.03 % | 93.87 % | 94.58 % | 93.84 % | 3.97 % | 6.02 % | 5.83 % | 10.23 |
| | GNB | 94.86 % | 92.67 % | 93.73 % | 92.95 % | 95.11 % | 92.52 % | 93.44 % | 92.45 % | 4.89 % | 7.33 % | 7.05 % | 17.46 |
| | SVM | 93.91 % | 90.95 % | 92.09 % | 91.23 % | 94.12 % | 91.18 % | 91.96 % | 90.79 % | 5.88 % | 9.05 % | 8.77 % | 23.65 |
| | BLR | 91.32 % | 87.56 % | 89.23 % | 88.12 % | 92.67 % | 87.98 % | 88.84 % | 87.23 % | 7.33 % | 12.44 % | 11.88 % | 35.78 |
| SQL_injection | EWEA | 97.81 % | 96.89 % | 97.35 % | 96.76 % | 98.14 % | 96.62 % | 97.18 % | 96.64 % | 1.86 % | 3.11 % | 3.24 % | 6.21 |
| | RF | 96.95 % | 95.56 % | 96.23 % | 95.78 % | 97.23 % | 95.48 % | 95.89 % | 95.42 % | 2.77 % | 4.44 % | 4.22 % | 9.34 |
| | DT | 95.98 % | 94.38 % | 95.13 % | 94.67 % | 96.42 % | 94.54 % | 94.92 % | 94.58 % | 3.58 % | 5.62 % | 5.33 % | 13.42 |
| | KNN | 95.12 % | 93.56 % | 94.42 % | 93.67 % | 95.21 % | 93.42 % | 93.78 % | 93.48 % | 4.79 % | 6.44 % | 6.33 % | 11.27 |
| | GNB | 94.23 % | 92.69 % | 93.32 % | 92.87 % | 94.67 % | 92.54 % | 92.89 % | 92.46 % | 5.33 % | 7.31 % | 7.13 % | 18.53 |
| | SVM | 93.37 % | 91.42 % | 92.54 % | 91.76 % | 93.83 % | 91.32 % | 91.98 % | 91.33 % | 6.17 % | 8.58 % | 8.24 % | 24.67 |
| | BLR | 91.72 % | 88.55 % | 90.01 % | 89.12 % | 92.37 % | 88.79 % | 89.68 % | 88.89 % | 7.63 % | 11.45 % | 10.88 % | 36.21 |
| Port_Scanning | EWEA | 98.63 % | 97.92 % | 98.27 % | 98.11 % | 99.02 % | 97.88 % | 98.13 % | 97.80 % | 0.98 % | 2.08 % | 1.89 % | 10.57 |
| | RF | 97.95 % | 96.58 % | 97.20 % | 96.87 % | 98.23 % | 96.71 % | 97.32 % | 96.74 % | 1.77 % | 3.42 % | 3.13 % | 15.89 |
| | DT | 97.13 % | 95.62 % | 96.45 % | 95.87 % | 97.42 % | 95.74 % | 96.01 % | 95.72 % | 2.58 % | 4.38 % | 4.13 % | 20.56 |
| | KNN | 96.48 % | 94.87 % | 95.52 % | 95.17 % | 96.71 % | 94.78 % | 95.12 % | 94.84 % | 3.29 % | 5.13 % | 4.83 % | 18.21 |
| | GNB | 95.71 % | 93.96 % | 94.71 % | 94.27 % | 95.91 % | 93.88 % | 94.13 % | 93.93 % | 4.09 % | 6.04 % | 5.73 % | 25.47 |
| | SVM | 94.89 % | 92.74 % | 93.62 % | 93.11 % | 94.67 % | 92.61 % | 92.98 % | 92.61 % | 5.33 % | 7.26 % | 6.89 % | 31.78 |
| | BLR | 92.35 % | 89.43 % | 90.98 % | 89.67 % | 93.21 % | 89.78 % | 90.23 % | 89.78 % | 6.79 % | 10.57 % | 10.33 % | 42.12 |
| XSS | EWEA | 98.21 % | 97.45 % | 97.86 % | 97.54 % | 98.78 % | 97.32 % | 97.98 % | 97.36 % | 1.22 % | 2.55 % | 2.46 % | 8.75 |
| | RF | 97.32 % | 96.06 % | 96.68 % | 96.28 % | 97.67 % | 95.94 % | 96.48 % | 96.01 % | 2.33 % | 3.94 % | 3.72 % | 13.45 |
| | DT | 96.45 % | 95.02 % | 95.72 % | 95.41 % | 96.83 % | 94.56 % | 95.12 % | 94.58 % | 3.17 % | 4.98 % | 4.59 % | 18.21 |
| | KNN | 95.67 % | 94.12 % | 94.89 % | 94.17 % | 95.98 % | 93.42 % | 94.76 % | 93.84 % | 4.02 % | 5.88 % | 5.83 % | 16.45 |
| | GNB | 94.87 % | 93.22 % | 93.97 % | 92.95 % | 94.99 % | 92.52 % | 93.78 % | 92.46 % | 5.01 % | 6.78 % | 7.05 % | 24.33 |
| | SVM | 94.01 % | 92.49 % | 92.78 % | 91.76 % | 93.64 % | 91.32 % | 92.56 % | 91.29 % | 6.36 % | 7.51 % | 8.24 % | 32.56 |
| | BLR | 92.13 % | 89.98 % | 90.87 % | 89.12 % | 92.99 % | 88.79 % | 89.98 % | 88.89 % | 7.01 % | 10.02 % | 10.88 % | 39.87 |
| MITM | EWEA | 99.15 % | 98.76 % | 98.95 % | 98.61 % | 99.57 % | 98.44 % | 98.56 % | 98.53 % | 0.43 % | 1.24 % | 1.39 % | 5.67 |

(*continued on next page*)

**Table 5** (*continued*)

| Attack type | Model | Accuracy | Precision | F1-score | Recall | Specificity | NPV | AUC | MCC | FPR | FDR | FNR | ADT(ms) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | RF | 98.27 % | 97.42 % | 97.84 % | 97.59 % | 98.71 % | 97.12 % | 97.68 % | 97.55 % | 1.29 % | 2.58 % | 2.41 % | 10.78 |
| | DT | 97.46 % | 96.59 % | 97.06 % | 96.87 % | 97.83 % | 96.74 % | 96.98 % | 96.89 % | 2.17 % | 3.41 % | 3.13 % | 15.34 |
| | KNN | 96.78 % | 95.82 % | 96.29 % | 95.97 % | 97.11 % | 95.62 % | 95.97 % | 95.86 % | 2.89 % | 4.18 % | 4.03 % | 13.98 |
| | GNB | 95.96 % | 94.87 % | 95.49 % | 94.97 % | 96.74 % | 94.78 % | 94.87 % | 94.76 % | 3.26 % | 5.13 % | 5.03 % | 21.45 |
| | SVM | 95.21 % | 93.95 % | 94.73 % | 94.17 % | 96.23 % | 93.42 % | 93.98 % | 93.84 % | 3.77 % | 6.05 % | 5.83 % | 28.56 |
| | BLR | 93.67 % | 91.82 % | 92.97 % | 91.98 % | 94.32 % | 91.78 % | 91.89 % | 91.67 % | 5.68 % | 8.18 % | 8.02 % | 36.89 |
| Fingerprinting | EWEA | 98.79 % | 97.92 % | 98.34 % | 98.11 % | 99.02 % | 97.88 % | 98.13 % | 97.80 % | 0.98 % | 2.08 % | 1.89 % | 11.56 |
| | RF | 97.91 % | 96.73 % | 97.30 % | 96.98 % | 98.08 % | 96.84 % | 97.32 % | 96.74 % | 1.48 % | 2.57 % | 2.02 % | 13.21 |
| | DT | 96.79 % | 95.46 % | 96.11 % | 95.66 % | 97.01 % | 95.62 % | 96.01 % | 95.72 % | 2.12 % | 3.01 % | 3.34 % | 18.45 |
| | KNN | 96.15 % | 94.86 % | 95.61 % | 95.11 % | 96.24 % | 95.03 % | 95.34 % | 94.84 % | 2.79 % | 4.15 % | 4.89 % | 15.67 |
| | GNB | 94.82 % | 93.37 % | 94.14 % | 93.41 % | 95.03 % | 93.75 % | 94.03 % | 93.93 % | 3.42 % | 5.64 % | 6.59 % | 22.56 |
| | SVM | 92.73 % | 90.82 % | 91.76 % | 91.24 % | 93.22 % | 91.45 % | 91.39 % | 90.98 % | 4.57 % | 7.38 % | 8.76 % | 31.89 |
| | BLR | 89.46 % | 86.93 % | 88.06 % | 87.28 % | 90.12 % | 87.40 % | 88.34 % | 87.92 % | 7.98 % | 11.94 % | 12.72 % | 39.56 |
| Uploading | EWEA | 99.62 % | 98.95 % | 99.24 % | 99.36 % | 99.75 % | 99.35 % | 99.11 % | 98.98 % | 0.25 % | 1.05 % | 0.64 % | 5.89 |
| | RF | 98.73 % | 97.84 % | 98.24 % | 97.98 % | 99.03 % | 97.58 % | 98.01 % | 97.79 % | 0.97 % | 2.16 % | 2.02 % | 12.34 |
| | DT | 97.85 % | 96.92 % | 97.37 % | 97.52 % | 98.21 % | 96.75 % | 97.32 % | 97.11 % | 1.79 % | 3.08 % | 2.48 % | 19.67 |
| | KNN | 97.23 % | 96.26 % | 96.86 % | 96.63 % | 97.82 % | 96.48 % | 96.89 % | 96.69 % | 2.18 % | 3.74 % | 3.37 % | 15.23 |
| | GNB | 96.42 % | 95.34 % | 95.85 % | 95.71 % | 96.91 % | 95.62 % | 95.89 % | 95.68 % | 3.09 % | 4.66 % | 4.29 % | 21.45 |
| | SVM | 95.56 % | 94.36 % | 94.87 % | 94.62 % | 96.02 % | 93.82 % | 94.45 % | 94.18 % | 3.98 % | 5.64 % | 5.38 % | 28.56 |
| | BLR | 92.87 % | 91.15 % | 91.96 % | 90.83 % | 93.68 % | 90.94 % | 90.87 % | 90.67 % | 6.32 % | 8.85 % | 9.17 % | 37.89 |
| Vuln_scanner | EWEA | 98.45 % | 97.82 % | 97.56 % | 97.41 % | 99.02 % | 97.32 % | 98.02 % | 97.39 % | 0.98 % | 2.18 % | 2.59 % | 12.56 |
| | RF | 97.68 % | 96.75 % | 97.09 % | 96.91 % | 98.12 % | 96.48 % | 97.32 % | 96.89 % | 1.88 % | 3.25 % | 3.09 % | 15.67 |
| | DT | 96.82 % | 95.71 % | 96.28 % | 95.86 % | 97.23 % | 95.62 % | 96.12 % | 95.67 % | 2.77 % | 4.29 % | 4.14 % | 21.34 |
| | KNN | 96.15 % | 94.86 % | 95.61 % | 95.11 % | 96.24 % | 95.03 % | 95.34 % | 94.84 % | 3.79 % | 4.15 % | 4.89 % | 17.45 |
| | GNB | 94.82 % | 93.37 % | 94.14 % | 93.41 % | 95.03 % | 93.75 % | 94.03 % | 93.93 % | 4.42 % | 5.64 % | 6.59 % | 24.56 |
| | SVM | 92.73 % | 90.82 % | 91.76 % | 91.24 % | 93.22 % | 91.45 % | 91.39 % | 90.98 % | 5.57 % | 7.38 % | 8.76 % | 32.89 |
| | BLR | 89.46 % | 86.93 % | 88.06 % | 87.28 % | 90.12 % | 87.40 % | 88.34 % | 87.92 % | 8.98 % | 11.94 % | 12.72 % | 39.56 |

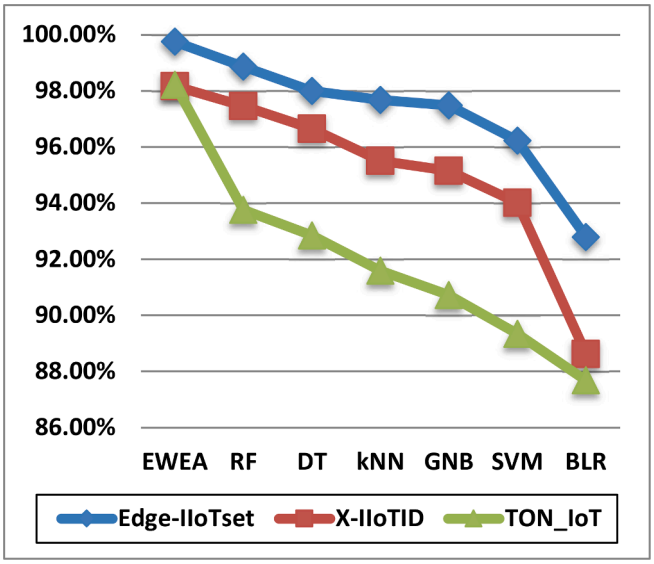



**Fig. 13.** Dataset dynamics: accuracy comparison across datasets.

network, showcasing the model's efficacy in reducing the impact and severity of detected attacks. The AME was determined through the sFlow monitoring dashboard. The mathematical expression is shown in Eq. (12)

$$AME = \frac{TNMA}{TNDA} \times 100 \tag{12}$$

**Area Under the Curve (AUC):** AUC, relevant to IoT intrusion detection, quantifies the overall discrimination ability of the model across different intrusion detection thresholds, contributing to an effective evaluation of its performance.

**Network Latency (NL):** NL, in the context of IoT intrusion detection, refers to the time delay experienced in communication within the network, including the processing time of the intrusion detection system.

In the context of Eqs. (1)–(12), false positives (FP) denote cases in which the system erroneously identifies an event as an attack. In contrast, false negatives (FN) indicate instances where an actual attack remains undetected. True positives (TP) and true negatives (TN) correspond to the accurate identification of attack and non-attack events, respectively. Moreover, TP, TN, FP, and FN serve as the foundation for constructing the confusion matrix depicted in Fig. 12. The term "SIDT" in Eq. (11) represents the time required to identify a single event during a test, while "TET" defines the overall number of experimental runs. In Eq. (12), the term "TNMA" denotes the total number of mitigated attacks, while "TNDA" signifies the total number of detected attacks.

Table 4 displays the assessment results for binary classification, while Table 5 displays the results of multi-class classification using the ML classifiers. Three different datasets—X-IIoTID, Edge-IIoTset, and TON_IoT—were used to evaluate these models. Overfitting was effectively mitigated by exposing the models to a variety of datasets, allowing for the assessment of their performance on previously unknown data as well as different types and circumstances of attacks. The models' capacity to identify and thwart different types of intrusion threats in SD-IoT networks was the main focus of the study.

Starting with the accuracy metric, the findings in Table 4 demonstrate that EWEA achieves the highest average accuracy rate of 99.75 % for the Edge-IIoTset dataset, 98.17 % for the TON-IoT dataset, and 98.21 % for the X-IIoTID dataset. It outperforms all other models across all datasets. With good accuracy rates, RF and DT come in close second and third, with 98.86 % and 97.99 % for Edge-IIoTset, 97.46 % and 96.65 % for TON-IoT, and 93.76 % and 92.84 % for X-IIoTID, respectively.

The performance of the other models varied across the datasets. KNN consistently showed good performance across all datasets, although its accuracy rate was slightly lower compared to RF and DT. GNB exhibited relatively lower accuracy rates across all datasets, indicating its limitations in accurately detecting attacks in diverse scenarios. Similarly, SVM and BLR demonstrated the lowest accuracy rates among the models, emphasizing the need for further improvement in their performance.

Fig. 13's results show that the Edge-IIoTset dataset had the best performance from the ML models. In comparison, the X-IIoTID dataset exhibited slightly lower accuracy across all models, indicating a potentially more challenging environment for intrusion detection. The TON_IoT dataset falls in between, demonstrating the models' effectiveness without reaching the high accuracy achieved in the Edge-IIoTset scenario. Overall, the comprehensive evaluation and comparison of the machine learning models underscore their effectiveness in diverse attack scenarios. The results emphasize the importance of dataset characteristics in influencing model performance, with the Edge-IIoTset dataset being particularly conducive to achieving robust and accurate intrusion detection.

Fig. 14 presents a visual comparison of accuracy, precision, F1, and recall. They assess how well the model performs in accurately detecting attack scenarios. EWEA maintains its superiority across all datasets. It achieves precision rates of 99.32 %, 97.63 %, and 98.87 % for Edge-IIoTset, TON_IoT, and X-IIoTID, respectively. Similarly, it attains F1 scores of 99.53 %, 97.90 %, and 97.54 %, and recall rates of 99.67 %, 98.08 %, and 98.98 % for the respective datasets. RF and DT consistently perform well in these metrics,

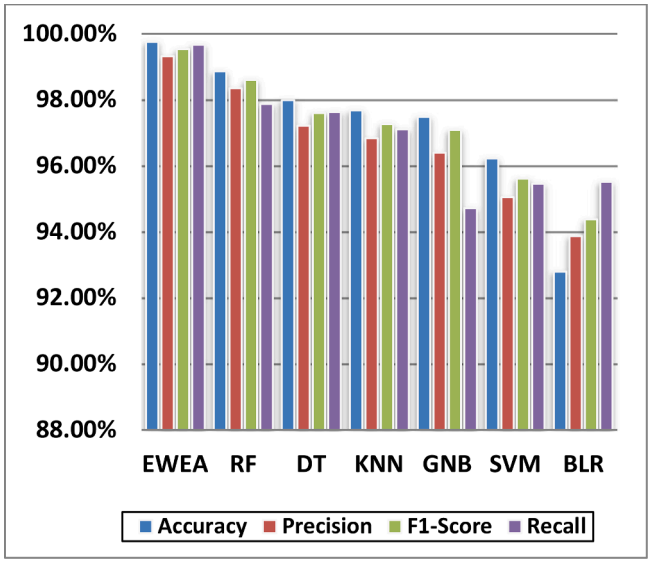



**Fig. 14.** Comparison of accuracy, precision, F1-score, and recall.

demonstrating their effectiveness in detecting attack instances. Analyzing TNR and NPV, which assess the model's ability to classify normal instances correctly, we observe that EWEA, RF, and DT exhibit high rates across all datasets. They consistently achieve TNR and NPV values above 96 %, indicating their proficiency in accurately identifying normal traffic.

Examining FPR, FDR, and FNR, which quantify the model's propensity for misclassifying normal instances as attacks, BLR exhibits the highest rates among the models across all datasets. It records FPR values of 6.92 %, 11.78 %, and 13.20 %, FDR values of 6.13 %, 11.76 %, and 14.66 %, and FNR values of 7.49 %, 11.88 %, and 13.26 % for Edge-IIoTset, TON_IoT, and X-IIoTID, respectively. These results suggest that BLR may struggle to differentiate between normal and attack instances accurately. Fig. 15 presents a visual comparison of FPR, FDR, and FNR.

Moving to MCC and AUC, which provide a comprehensive evaluation of the models' performance, EWEA demonstrates the highest values among all models for all datasets. It achieves MCC scores of 99.14 %, 96.31 %, and 95.82 %, and AUC values of 99.83 %, 98.03 %, and 97.68 % for Edge-IIoTset, TON_IoT, and X-IIoTID, respectively. These results confirm the robustness and effectiveness of EWEA in accurately detecting attacks across different datasets. Fig. 16 presents a visual comparison of AUC and MCC.

We looked beyond just accuracy to assess how well the models maintained their effectiveness under changing network conditions. This stability is measured by the Average Accuracy Stability Rate (AASR). We tested the models with varying traffic densities (from

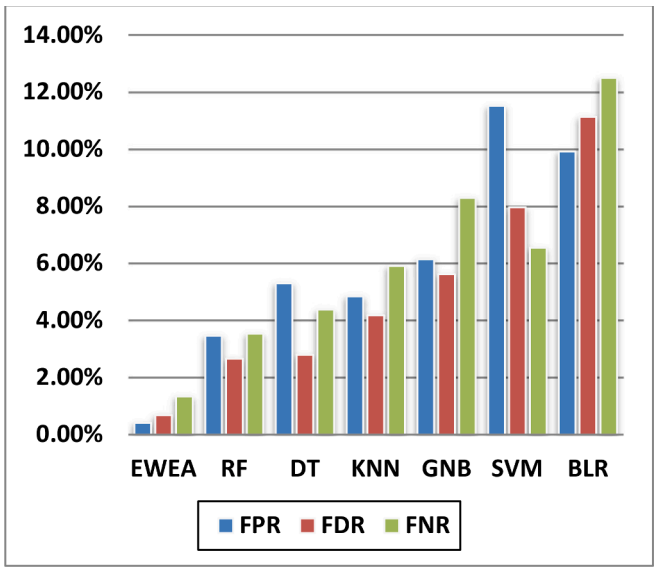



**Fig. 15.** Comparison of FPR, FDR, and FNR.

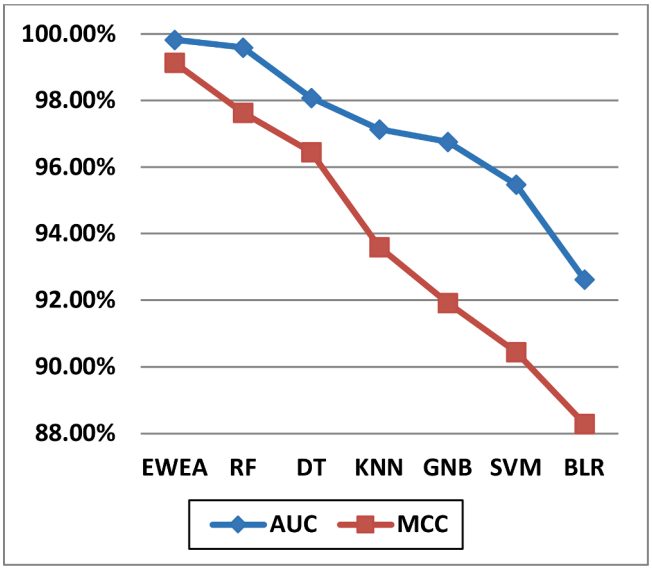



**Fig. 16.** Comparison of AUC and MCC.

5000 to 50,000 packets) and different attack rates (20–80 % of total traffic).

Fig. 17 visually shows the AASR of the seven Machine Learning models. The plot reveals that some models are more stable than others when facing changes in traffic and attack intensity. For example, BLR and SVM had the lowest AASR. While their accuracy initially rose with higher traffic, it ultimately decreased. Specifically, BLR achieved 92.79 % accuracy at a low traffic density (5000 packets) and a mild attack rate (20 %). However, this accuracy improved to 93.87 % at moderate traffic (35,000 packets) with a slightly higher attack rate (30 %). Unfortunately, at the highest traffic level (50,000 packets) with a moderate attack rate (40 %), the accuracy dropped to 90.63 %.

Analogously, SVM demonstrated a 95.22 % accuracy rate at a 5000 flow rate and a 20 % attack rate, which increased to 97.04 % at a flow rate of 45,000 but dropped to 93.07 % at a flow rate of 50,000 and an attack rate of 50 %. The inconsistent detection accuracy with varying flow or attack rates could negatively impact the models' performance during high flow or attack intensities, particularly in scenarios like DDoS attacks. On the other hand, GNB and KNN displayed similar performance patterns, while DT and RF outperformed the other models except for EWEA. The EWEA model demonstrated an accuracy of 95.79 % at a flow rate of 5000 and an attack rate of 20 %, and it consistently increased to 99.89 % at a flow rate of 50,000 and an attack rate of 40 %. These findings indicate that EWEA outperformed other models in terms of accuracy and consistency across different flow rates and attack rates.

Fig. 18 focuses on Average Detection Time (ADT), which measures how quickly the models identify attacks. The proposed framework, EWEA, consistently outperforms others in all datasets, achieving an ADT of 2.11 ms for Edge-IIoTset, 14.22 ms for TON-IoT, and 18.67 ms for X-IIoTID. These rapid detection times contribute to the framework's ability to respond to threats in real-time. Regarding Network Latency (NL), the proposed framework again delivers impressive results. NL values range from 3.34 ms to 25.72 ms, indicating minimal impact on network responsiveness. This ensures efficient operation without compromising network performance.

Furthermore, the Attack Mitigation Effectiveness (AME) is exceptionally high, ranging from 88.29 % to 99.95 %. This signifies the framework's success in effectively counteracting and neutralizing attacks, safeguarding network integrity. Overall, the experiments

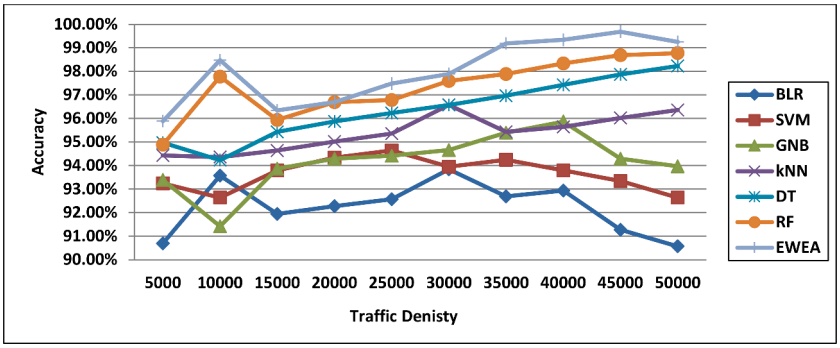



**Fig. 17.** Stability in accuracy - visualizing the average accuracy stability rate.

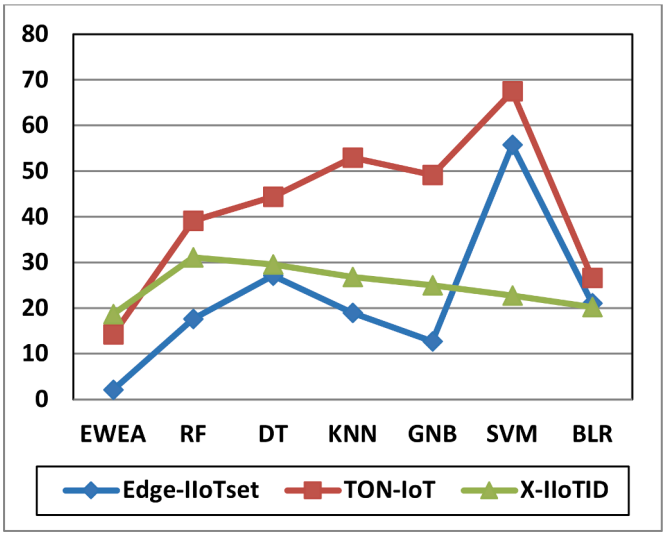



**Fig. 18.** Comparison of the ADT (ms) for the datasets.

confirm the efficiency of the proposed approach in identifying and thwarting intrusion threats in SDN-enabled IoT networks. The combination of high accuracy, low false positive rates, rapid detection times, excellent AME, and minimal NL establishes this framework as a reliable and efficient solution for network security and optimal performance. Fig. 19 visually compares the accuracy of various Machine Learning and Deep Learning methods.

## 5. Comparative analysis: current methods vs. the multi-block framework

Our research introduces the MULTI-BLOCK framework for intrusion detection in SD-IoT networks. This section compares MULTI-BLOCK to existing methods, highlighting its advantages based on various criteria. The proposed framework achieves exceptional performance compared to existing approaches. While some methods boast high accuracy, they often come with a trade-off in terms of false positives (FPR). MULTI-BLOCK strikes a balance, achieving a mean precision of 99.75 % with a remarkably low FPR of 0.40 %. This translates to highly accurate intrusion detection with minimal false alarms, reducing unnecessary network disruptions.

Unlike methods limited to specific datasets or attack types, MULTI-BLOCK leverages diverse real-world datasets like X-IIoTID, TON_IoT, and Edge-IIoTset. This comprehensive data utilization ensures the framework's effectiveness against a wider range of intrusion scenarios commonly encountered in SD-IoT networks. Additionally, MULTI-BLOCK successfully identifies 13 distinct intrusion attack types, showcasing its adaptability to diverse network environments and attack patterns.

Several existing methods rely on simulated or outdated datasets, potentially hindering their effectiveness against evolving attack strategies. A key strength of MULTI-BLOCK is its utilization of real IoT traffic features for crafting the detection model. This approach offers a significant advantage in terms of generalizability and applicability to real-world situations.

The MULTI-BLOCK framework employs a modular architecture with a pyramidal conceptually decentralized multi-controller structure. This design, combined with P4-enabled switches, fosters scalability and resilience. Compared to methods with limited scalability, MULTI-BLOCK can effectively handle complex and coordinated attacks across large-scale SD-IoT networks.

Table 6 provides a detailed comparison between the proposed framework and existing methods, highlighting key metrics such as dataset usage, attack type coverage, scalability, FPR, and accuracy. This table allows for a quick visual comparison of MULTI-BLOCK's strengths. By addressing the limitations of existing methods, MULTI-BLOCK offers a comprehensive and adaptable solution for intrusion detection in SD-IoT networks. Its low FPR, high accuracy, robust feature set, and real-world data foundation ensure effective protection against a wide range of intrusion threats. Additionally, the framework's scalable and decentralized architecture allows it to handle complex attacks in large-scale SD-IoT environments.

## 6. Limitations and future works

The MULTI-BLOCK framework demonstrates a significant leap forward in securing IoT networks against intrusion attacks. However, continuous improvement is essential to ensure its long-term effectiveness. One key limitation to consider is the potential for increased computational overhead. While P4 technology facilitates efficient traffic processing, the combined use of stateful processing and machine learning algorithms could introduce a burden on resource-constrained IoT devices. Future work should explore techniques for optimizing resource consumption within the framework. This will ensure its suitability for deployment in diverse IoT environments with varying device capabilities.

While emulation and simulation environments like Mininet-Wifi provide valuable insights, real-world testing is crucial. Future work should involve testing the framework's effectiveness against various intrusion attempts using real or virtual machines infected

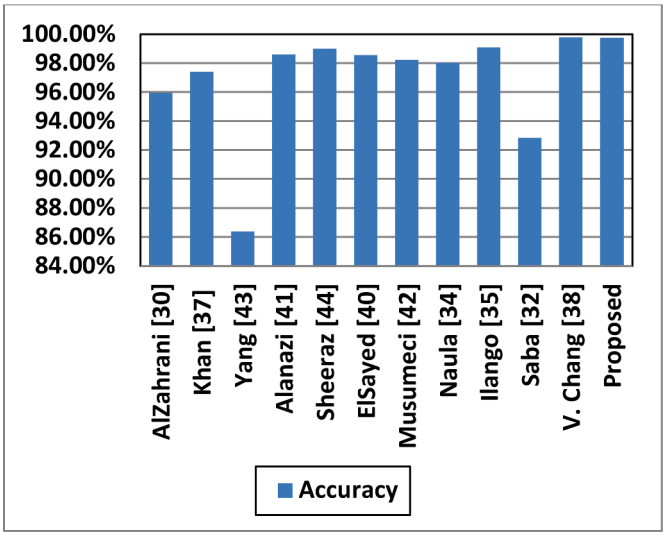



**Fig. 19.** Assessing accuracy against recent methods.

**Table 6**
Framework face-off - comparison between the proposed framework and current methods.

| Author | Year | Technique | Dataset | Multi-class? | Threat | Scalable? | Infrastructure | FPR | Accuracy |
|---|---|---|---|---|---|---|---|---|---|
| V. Chang [38] | 2020 | Deep-IFS: Deep Learning model for IIoT intrusion detection | Bot-IIoT | Yes | Intrusion attacks | Yes | IIoT | 1.8 % | 99.77 % |
| AlZahrani [30] | 2021 | DT, RF and XGBoost | NSL-KDD | Yes | Four intrusion types | No | SDN | – | 95.95 % |
| ElSayed [40] | 2021 | Hybrid model: CNN and a regularization technique | InSDN dataset | Yes | Intrusion attacks | No | SDN | – | 98.54 % |
| Alanazi [41] | 2022 | Set of ML Classifiers | X-IIoTID | Yes | Intrusion attacks | – | IoT | 00.50 % | 98.59 % |
| Musumeci[42] | 2022 | Set of ML Classifiers | Simulated | No | DDoS-TCP flood | No | SDN | 03.04 % | 98.22 % |
| Naula [34] | 2022 | Deep reinforcement learning | Simulated | No | LR-DDoS | – | SDN | – | 98.02 % |
| Ilango [35] | 2022 | Feed Forward CNN-based | CIC-2017 | No | LR-DDoS | – | IoT | – | 99.09 % |
| Saba [32] | 2022 | An anomaly-based detection method using CNN | BoT-IoT | Yes | Intrusion attacks | No | IoT | – | 92.85 % |
| Khan [37] | 2023 | Ensemble model using soft voting | CIC-2017 | Yes | Intrusion attacks | No | IIoT | 03.78 % | 97.40 % |
| Yang [43] | 2023 | Ensemble learning technique | N-BaIoT dataset | Yes | Rare-Class Intrusion | – | IoT | – | 86.39 % |
| Sheeraz [44] | 2023 | MLP-based classification algorithm | CTU-dataset | No | DDoS attack | No | Conventional | 02.11 % | 98.99 % |
| Nandanwar et al. [39] | 2024 | Adaptive-based CNN-GRU for botnet detection in IIoT | N-BaIoT dataset | Yes | Botnet attacks | No | IIoT | – | 99.75 % |
| Proposed | 2024 | Enhanced Weighted Ensemble Voting Algorithm | TON_IoT, and Edge-IIoTset | Yes | 13 intrusion attack types | Yes | SD-IoT | 0.40 % | 99.75 % |

with viruses, trojans, and other malware. This will provide a more comprehensive evaluation of its ability to detect and mitigate real-world threats.

Another area for exploration is scalability in extremely large networks. The decentralized architecture of MULTI-BLOCK inherently promotes scalability. However, its effectiveness in real-world scenarios with a vast number of devices needs further evaluation. Extensive testing within large-scale virtual machine environments, alongside real-world testing, can provide valuable insights into the framework's ability to handle extensive deployments. This comprehensive testing approach will solidify our understanding of MULTI-BLOCK's scalability and pave the way for its implementation in large-scale IoT networks.

While the achieved False Positive Rate (FPR) is promising, there's always room for improvement. Future research can investigate techniques to reduce false positives further. Minimizing these false positives will not only enhance the overall efficiency of the framework but also reduce the number of unnecessary mitigation actions triggered by the system. Additionally, exploring dynamic adaptive control mechanisms and enhanced mitigation strategies holds promise for improving the framework's adaptability and

effectiveness. By incorporating these advancements, MULTI-BLOCK can become even more adept at handling evolving threats and ensuring the security of IoT networks.

Finally, continuous monitoring and adaptation are crucial for addressing new attack vectors that may emerge. The framework is currently designed to identify a diverse range of attack types. However, new and unforeseen threats can always develop. By implementing ongoing monitoring and incorporating mechanisms for adaptation, MULTI-BLOCK can maintain its effectiveness in the face of a constantly evolving threat landscape. Through these future research directions, we can solidify MULTI-BLOCK's position as a robust and adaptable security solution for IoT networks.

## 7. Conclusion

The ever-expanding realm of IoT networks presents significant security challenges. Traditional approaches often struggle with efficient traffic monitoring, particularly when aiming to minimize synchronization and communication overhead. The heterogeneous nature of IoT devices, coupled with diverse communication protocols and often inadequate security measures, creates a fertile ground for sophisticated intrusion attempts. The potential for compromised devices to fuel large-scale botnets and DDoS attacks further intensifies these concerns. Current defences also fall short in the face of coordinated and multi-targeted attacks, highlighting the urgent need for innovative intrusion detection solutions. Furthermore, the limited availability of research utilizing real-world IoT network traffic data hinders the development of practical intrusion detection systems tailored to the unique challenges of these networks. Finally, many existing solutions are computationally expensive and time-consuming, making them unsuitable for resource-constrained IoT environments.

The MULTI-BLOCK framework directly addresses these multifaceted issues by proposing a novel, multi-module intrusion detection framework. This work contributes significantly to the field through several key advancements. One key innovation lies in the introduction of a novel set of 30 P4-extracted/calculated features. This enhanced feature set improves the accuracy of attack identification, enabling the framework to distinguish between legitimate traffic and malicious activities effectively. Additionally, the PCDMCS module facilitates real-time communication among controllers and switches. This minimizes delays in threat detection and containment, bolstering the overall security posture of the network.

For high-precision and swift attack identification, P4-DIDPS takes advantage of both stateful P4 processing and the EWEA. This proactive approach allows for swift mitigation actions, minimizing potential damage caused by intrusions. The P4-DAMS, a decentralized approach to attack mitigation, ensures scalability and efficient network resource utilization. This is particularly crucial in large-scale IoT deployments where centralized mitigation strategies might become unwieldy. Extensive evaluation demonstrates MULTI-BLOCK's effectiveness. The framework achieves outstanding results across 14 metrics, including accuracy, precision, recall, F1-score, and a remarkably low False Positive Rate. Additionally, the framework demonstrates stability under varying network conditions through the introduction of the AASR metric.

By combining these strengths, MULTI-BLOCK represents a significant advancement in IoT network security. It offers a robust and adaptable solution against evolving threats, paving the way for more secure and resilient IoT deployments. However, continuous research and refinement are crucial for maintaining effectiveness in the face of an ever-changing threat landscape. This ongoing pursuit will ensure the continued security and integrity of IoT networks in the years to come.

## CRediT authorship contribution statement

**Ahmed A. Toony:** Writing – review & editing, Methodology, Conceptualization. **Fayez Alqahtani:** Writing – review & editing, Methodology, Conceptualization. **Yasser Alginahi:** Writing – review & editing, Methodology, Conceptualization. **Wael Said:** Writing – original draft, Validation, Software, Methodology, Investigation, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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## References

[1] S.A. Bakhsh, M.A. Khan, F. Ahmed, M.S. Alshehri, H. Ali, J. Ahmad, Enhancing IoT network security through deep learning-powered intrusion detection system, Internet Things 24 (2023) 100936.

[2] S. Fraihat, S. Makhadmeh, M. Awad, M.A. Al-Betar, A. Al-Redhaei, Intrusion detection system for large-scale IoT NetFlow networks using machine learning with modified arithmetic optimization algorithm, Internet Things (2023) 100819.
[3] V. Chang, L. Golightly, P. Modesti, Q.A. Xu, L.M.T. Doan, K. Hall, et al., A survey on intrusion detection systems for fog and cloud computing, Future Internet 14 (2022) 89.
[4] V. Mothukuri, P. Khare, R.M. Parizi, S. Pouriyeh, A. Dehghantanha, G. Srivastava, Federated-learning-based anomaly detection for IoT security attacks, IEEE Internet Things J. 9 (2021) 2545–2554, https://doi.org/10.1109/JIOT.2021.3077803.
[5] A. Adeel, M. Gogate, S. Farooq, C. Ieracitano, K. Dashtipour, H. Larijani, et al., A survey on the role of wireless sensor networks and IoT in disaster management. Geological Disaster Monitoring Based on Sensor Networks, 2019, pp. 57–66.
[6] R. Ahmad, I. Alsmadi, Machine learning approaches to IoT security: a systematic literature review, Internet Things 14 (2021) 100365, https://doi.org/10.1016/j.iot.2021.100365.
[7] T. Gaber, J.B. Awotunde, M. Torky, S.A. Ajagbe, M. Hammoudeh, W. Li, Metaverse-IDS: deep learning-based intrusion detection system for metaverse-IoT networks, Internet Things 24 (2023) 100977.
[8] B. Kaur, S. Dadkhah, F. Shoeleh, E.C.P. Neto, P. Xiong, S. Iqbal, et al., Internet of things (IoT) security dataset evolution: challenges and future directions, Internet Things (2023) 100780.
[9] W.I. Khedr, A.E. Gouda, E.R. Mohamed, FMDADM: a multi-layer DDos attack detection and mitigation framework using machine learning for stateful SDN-based IoT networks, IEEE Access. 11 (2023) 28934–28954.
[10] A. Jaszcz, D. Połap, AIMM: artificial intelligence merged methods for flood DDoS attacks detection, J. King Saud Univ.-Comput. Inf. Sci. 34 (2022) 8090–8101.
[11] X.-H. Nguyen, K.-H. Le, Robust detection of unknown DoS/DDoS attacks in IoT networks using a hybrid learning model, Internet Things 23 (2023) 100851.
[12] X. Zhao, H. Su, Z. Sun, An intrusion detection system based on genetic algorithm for software-defined networks, Mathematics 10 (2022) 3941.
[13] M. Kumar, P. Mukherjee, S. Verma, Kavita, N. Jhanjhi, S. Vimal, et al., A novel SDN-based security framework for wireless sensor networks using TDCNN and PGF-ECC, Hum.-centric Comput. Inf. Sci. 13 (2023).
[14] J. Kim, Y. Kim, V. Yegneswaran, P. Porras, S. Shin, T. Park, Extended data plane architecture for in-network security services in software-defined networks, Comput. Secur. 124 (2023) 102976.
[15] I.H. Abdulqadder, D. Zou, I.T. Aziz, The DAG blockchain: a secure edge assisted honeypot for attack detection and multi-controller based load balancing in SDN 5G, Future Gener. Comput. Syst. 141 (2023) 339–354.
[16] B. Isyaku, K.B.A. Bakar, F.A. Ghaleb, A. Al-Nahari, Dynamic routing and failure recovery approaches for efficient resource utilization in OpenFlow-SDN: a survey, IEEe Access 10 (2022) 121791–121815, https://doi.org/10.1109/ACCESS.2022.3222849.
[17] F. Paolucci, F. Cugini, P. Castoldi, T. Osiński, Enhancing 5G SDN/NFV edge with P4 data plane programmability, IEEE Netw. 35 (2021) 154–160, https://doi.org/10.1109/MNET.021.1900599.
[18] X. Zhang, L. Cui, K. Wei, F.P. Tso, Y. Ji, W. Jia, A survey on stateful data plane in software defined networks, Comput. Netw. 184 (2021) 107597, https://doi.org/10.1016/j.comnet.2020.107597.
[19] A. Mahmood, C. Casetti, C.F. Chiasserini, P. Giaccone, J. Härri, Efficient caching through stateful SDN in named data networking, Trans. Emerg. Telecommun. Technol. 29 (2018) e3271, https://doi.org/10.1002/ett.3271.
[20] S. Kaur, K. Kumar, N. Aggarwal, A review on P4-Programmable data planes: architecture, research efforts, and future directions, Comput. Commun. 170 (2021) 109–129.
[21] L. Chen, Z. Wang, R. Huo, T. Huang, An adversarial DBN-LSTM method for detecting and defending against DDoS attacks in SDN environments, Algorithms 16 (2023) 197.
[22] F. Hauser, M. Häberle, D. Merling, S. Lindner, V. Gurevich, F. Zeiger, et al., A survey on data plane programming with p4: fundamentals, advances, and applied research, J. Netw. Comput. Appl. 212 (2023) 103561.
[23] W.I. Khedr, A.E. Gouda, E.R. Mohamed, P4-HLDMC: a novel framework for DDoS and ARP attack detection and mitigation in SD-IoT networks using machine learning, stateful P4, and distributed multi-controller architecture, Mathematics 11 (2023) 3552.
[24] S.F. Abbasi, H. Jamil, W. Chen, EEG-based neonatal sleep stage classification using ensemble learning, Comput. Mater. Contin. 70 (2022) 4619–4633.
[25] Z. Liu, Y. Wang, F. Feng, Y. Liu, Z. Li, Y. Shan, A DDoS detection method based on feature engineering and machine learning in software-defined networks, Sensors 23 (2023) 6176.
[26] I.K. Thajeel, K. Samsudin, S.J. Hashim, F. Hashim, Dynamic feature selection model for adaptive cross site scripting attack detection using developed multi-agent deep Q learning model, J. King Saud Univ.-Comput. Inf. Sci. 35 (2023) 101490.
[27] H. Elubeyd, D. Yiltas-Kaplan, Hybrid deep learning approach for automatic DoS/DDoS attacks detection in software-defined networks, Appl. Sci. 13 (2023) 3828.
[28] Z. Long, W. Jinsong, A hybrid method of entropy and SSAE-SVM based DDoS detection and mitigation mechanism in SDN, Comput. Secur. 115 (2022) 102604.
[29] N. Ravi, S.M. Shalinie, Learning-driven detection and mitigation of DDoS attack in IoT via SDN-cloud architecture, IEEE Internet Things J. 7 (2020) 3559–3570, https://doi.org/10.1109/JIOT.2020.2973176.
[30] A.O. Alzahrani, M.J. Alenazi, Designing a network intrusion detection system based on machine learning for software defined networks, Future Internet 13 (2021) 111.
[31] O. Yousuf, R.N. Mir, DDoS attack detection in Internet of Things using recurrent neural network, Comput. Electr. Eng. 101 (2022) 108034, https://doi.org/10.1016/j.compeleceng.2022.108034.
[32] T. Saba, A. Rehman, T. Sadad, H. Kolivand, S.A. Bahaj, Anomaly-based intrusion detection system for IoT networks through deep learning model, Comput. Electr. Eng. 99 (2022) 107810.
[33] W.I. Khedr, A.E. Gouda, E.R. Mohamed, FMDADM: a multi-layer DDoS attack detection and mitigation framework using machine learning for stateful SDN-based IoT networks, IEEE Access (2023).
[34] N.M. Yungaicela-Naula, C. Vargas-Rosales, J.A. Pérez-Díaz, D.F. Carrera, A flexible SDN-based framework for slow-rate DDoS attack mitigation by using deep reinforcement learning, J. Netw. Comput. Appl. 205 (2022) 103444.
[35] H.S. Ilango, M. Ma, R. Su, A FeedForward–Convolutional Neural Network to Detect Low-Rate DoS in IoT, Eng. Appl. Artif. Intell. 114 (2022) 105059, https://doi.org/10.1016/j.engappai.2022.105059.
[36] M.V. de Assis, L.F. Carvalho, J.J. Rodrigues, J. Lloret, M.L. Proença Jr, Near real-time security system applied to SDN environments in IoT networks using convolutional neural network, Comput. Electr. Eng. 86 (2020) 106738, https://doi.org/10.1016/j.compeleceng.2020.106738.
[37] M.A. Khan, N. Iqbal, H. Jamil, D.-H. Kim, An optimized ensemble prediction model using AutoML based on soft voting classifier for network intrusion detection, J. Netwo. Comput. Appl. 212 (2023) 103560.
[38] M. Abdel-Basset, V. Chang, H. Hawash, R.K. Chakrabortty, M. Ryan, Deep-IFS: intrusion detection approach for industrial internet of things traffic in fog environment, IEEE Trans. Industr. Inform. 17 (2020) 7704–7715.
[39] H. Nandanwar, R. Katarya, Deep learning enabled intrusion detection system for Industrial IOT environment, Expert Syst. Appl. 249 (2024) 123808.
[40] M.S. ElSayed, N.-A. Le-Khac, M.A. Albahar, A. Jurcut, A novel hybrid model for intrusion detection systems in SDNs based on CNN and a new regularization technique, J. Netw. Comput. Appl. 191 (2021) 103160.
[41] M. Alanazi, A. Aljuhani, Anomaly detection for internet of things cyberattacks, CMC-Comput. Mater. Contin. 72 (2022) 261–279, https://doi.org/10.32604/cmc.2022.024496.
[42] F. Musumeci, A.C. Fidanci, F. Paolucci, F. Cugini, M. Tornatore, Machine-learning-enabled DDoS attacks detection in P4 programmable networks, J. Netw. Syst. Manag. 30 (2022) 1–27.
[43] Y. Yang, Y. Gu, Y. Yan, Machine learning-based intrusion detection for rare-class network attacks, Electronics (Basel) 12 (2023) 3911.
[44] S. Ahmed, Z.A. Khan, S.M. Mohsin, S. Latif, S. Aslam, H. Mujlid, et al., Effective and efficient DDoS attack detection using deep learning algorithm, multi-layer perceptron, Future Internet 15 (2023) 76.